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Jenkins et al.

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(54) **SYMBOL DISPLAY ACCUMULATION
SEQUENCE AND A RELATED
ACCUMULATED SYMBOL DISPLAY USE
SEQUENCE**

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CPC **G07F 17/3213** (2013.01); **G07F 17/3267**
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See application file for complete search history.

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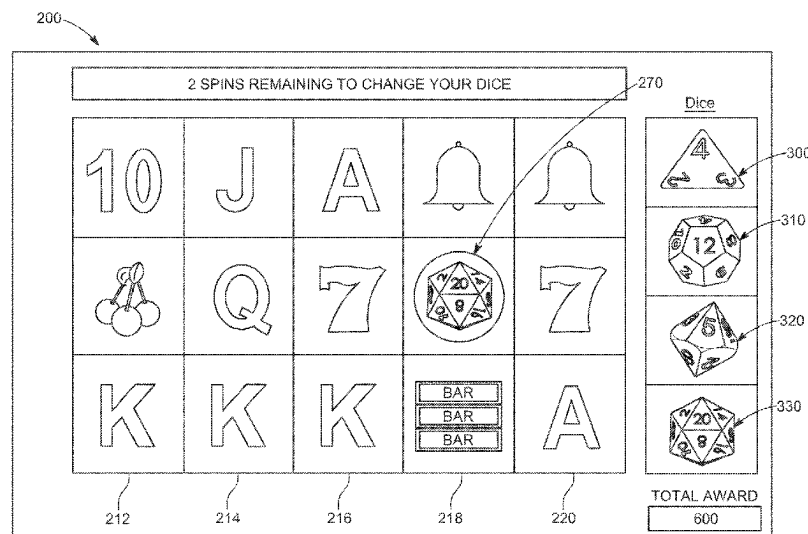
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(57) **ABSTRACT**

Gaming systems and methods that provide a symbol display
accumulation sequence and a related accumulated symbol
display use sequence, wherein the accumulation sequence
includes an accumulation of symbol displays during one or
more plays of one or more games during, and wherein the
use sequence includes the use of the accumulated symbol
displays.

19 Claims, 18 Drawing Sheets



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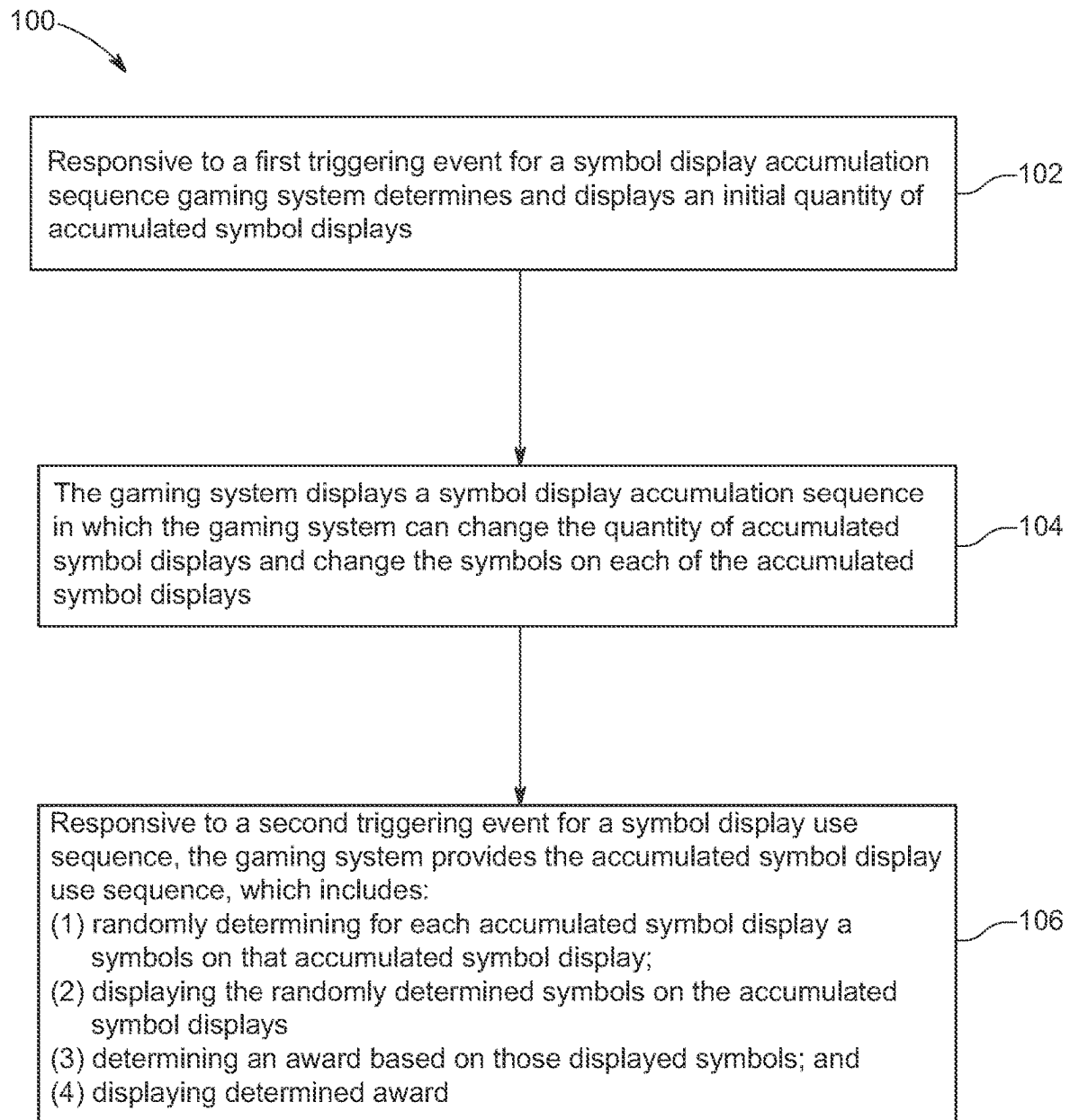


FIG. 1

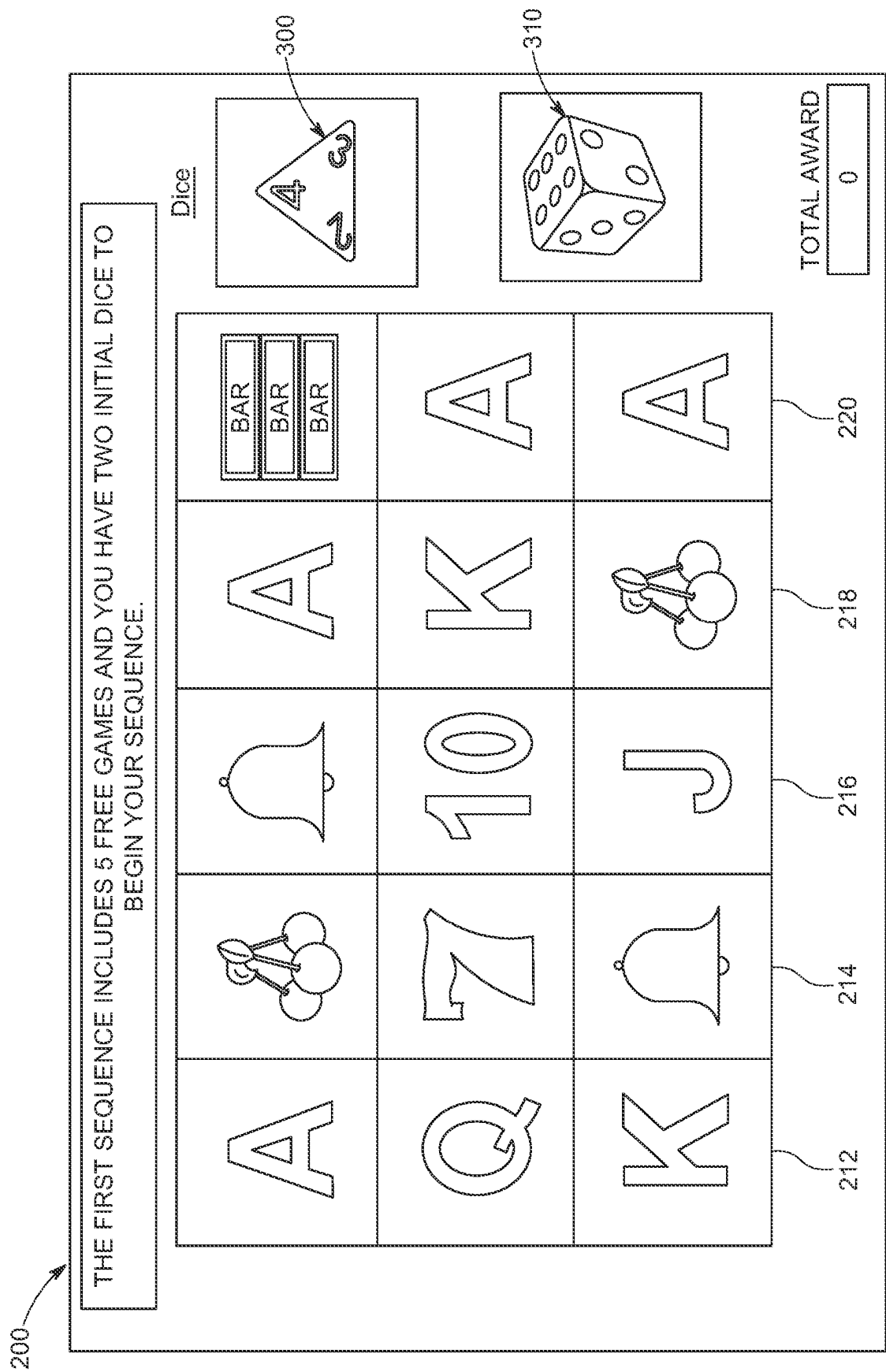


FIG. 2A

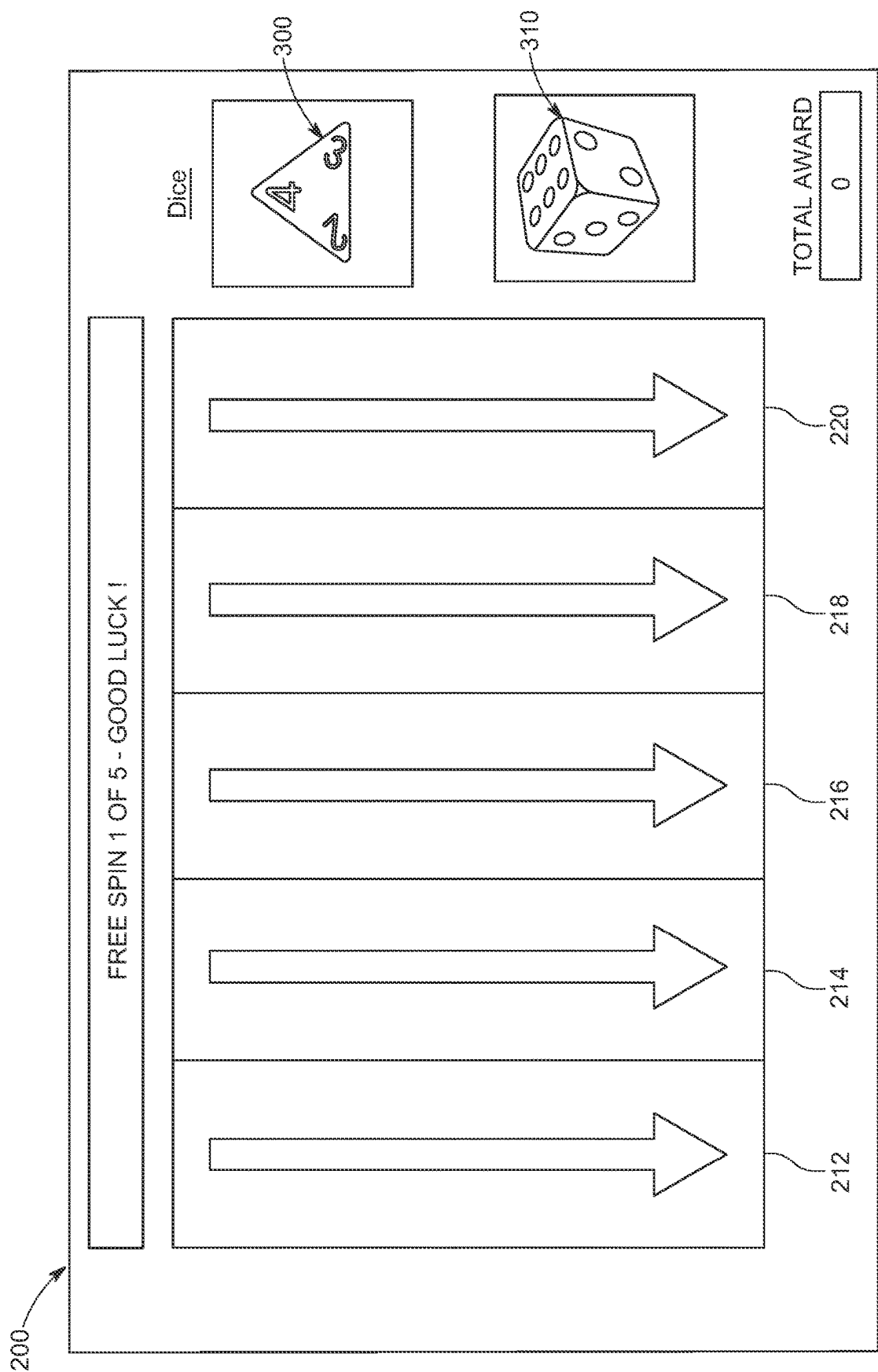


FIG. 2B

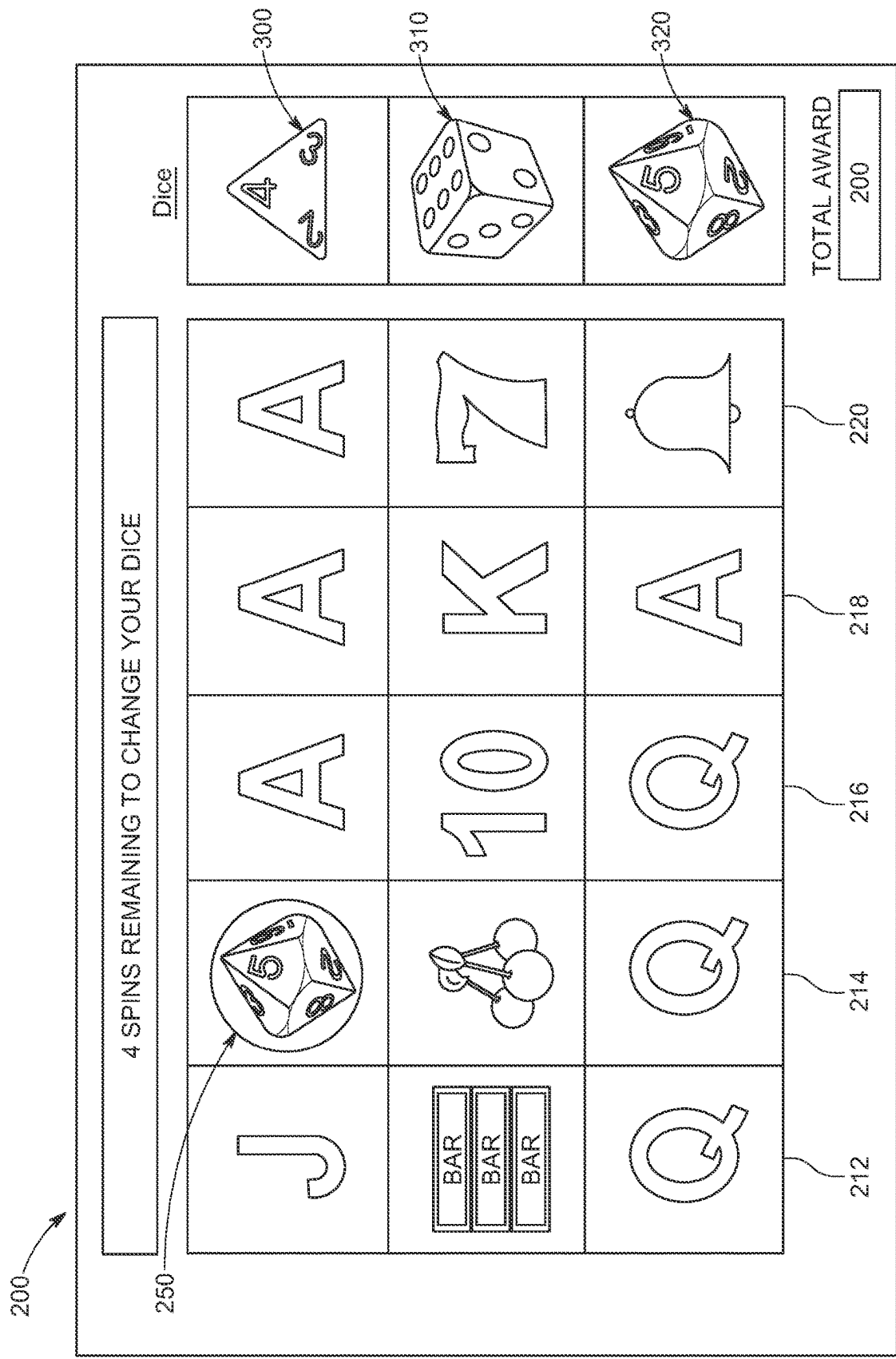


FIG. 2C

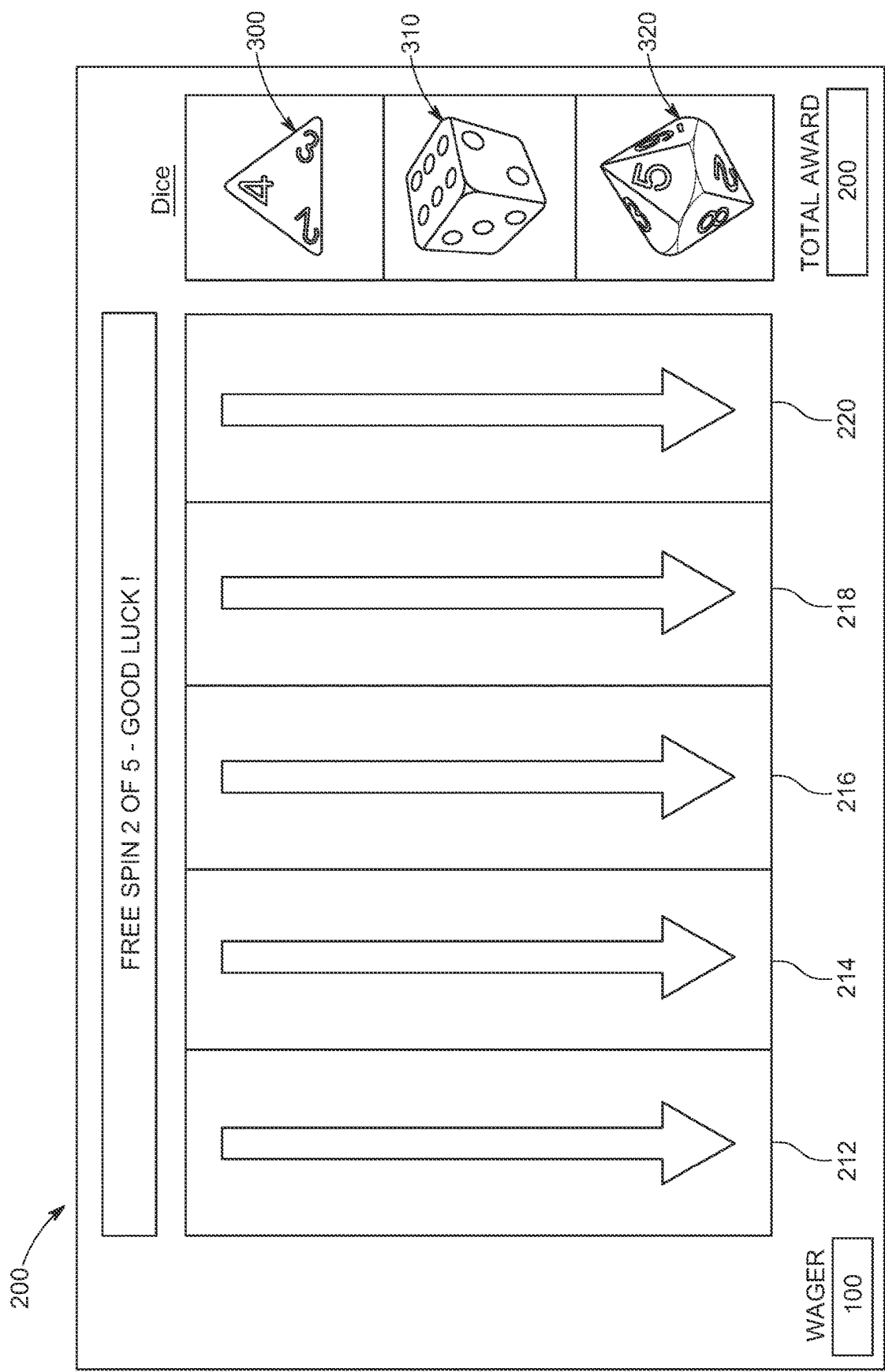


FIG. 2D

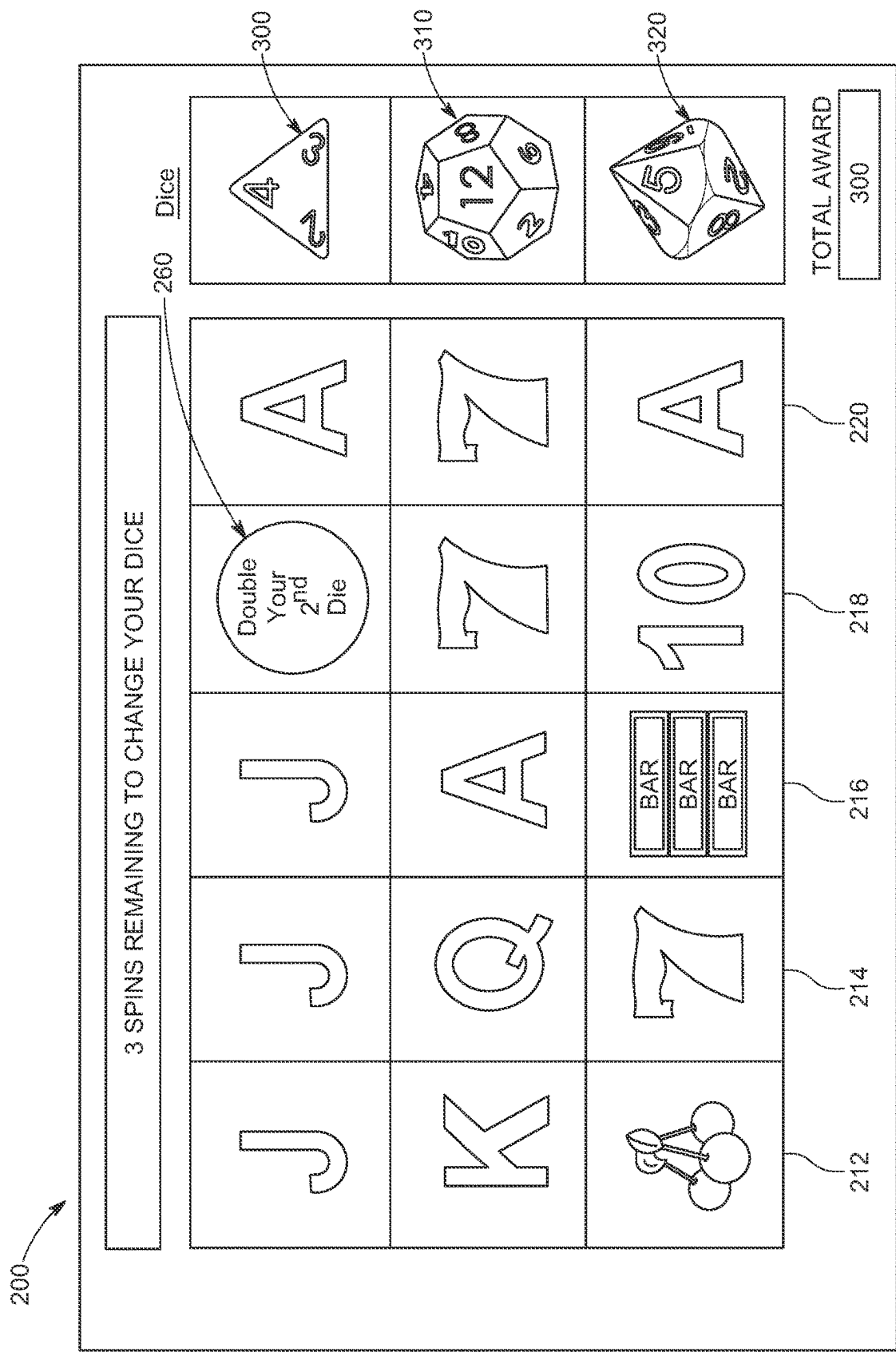


FIG. 2E

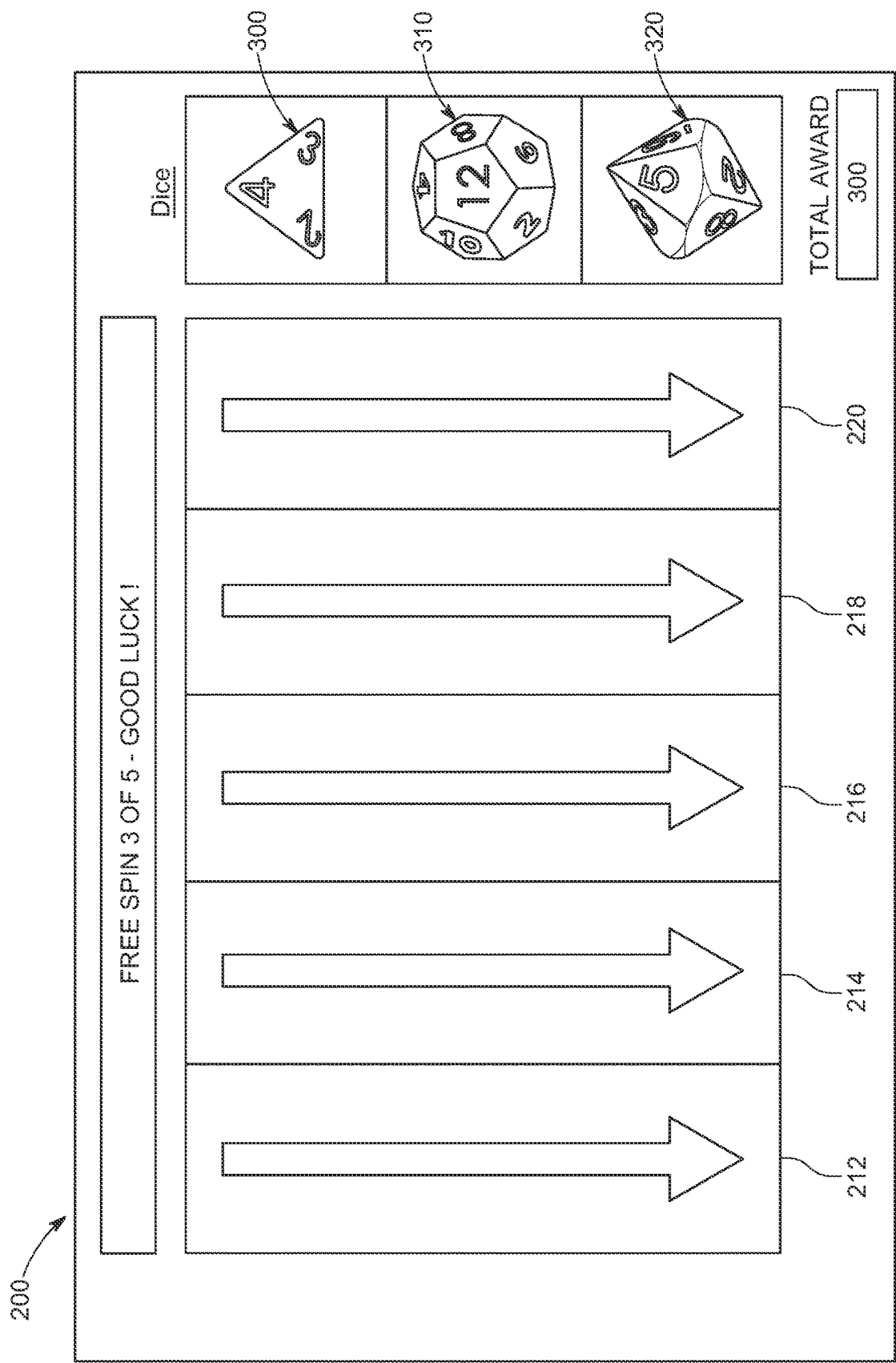


FIG. 2F

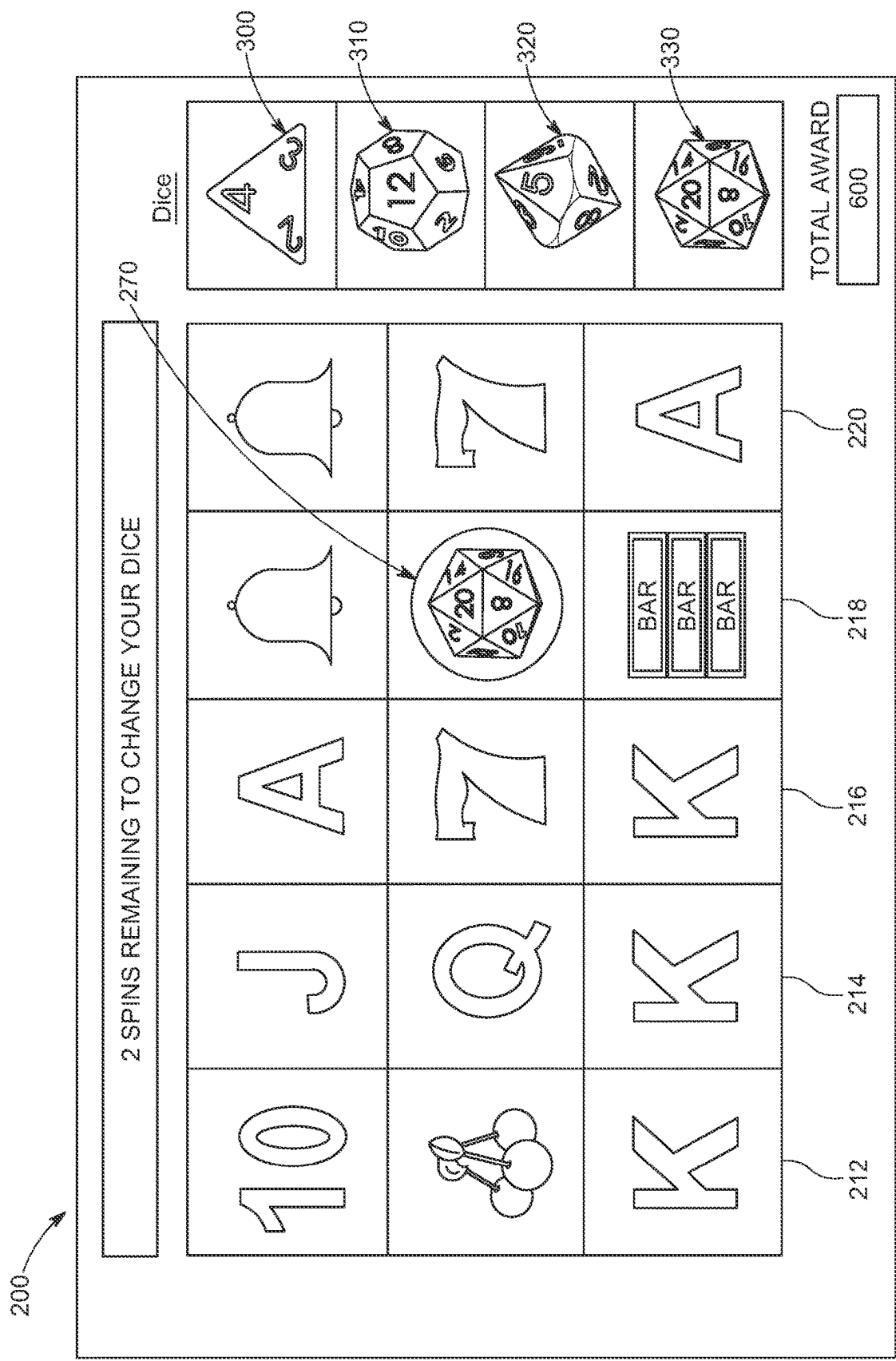


FIG. 2G

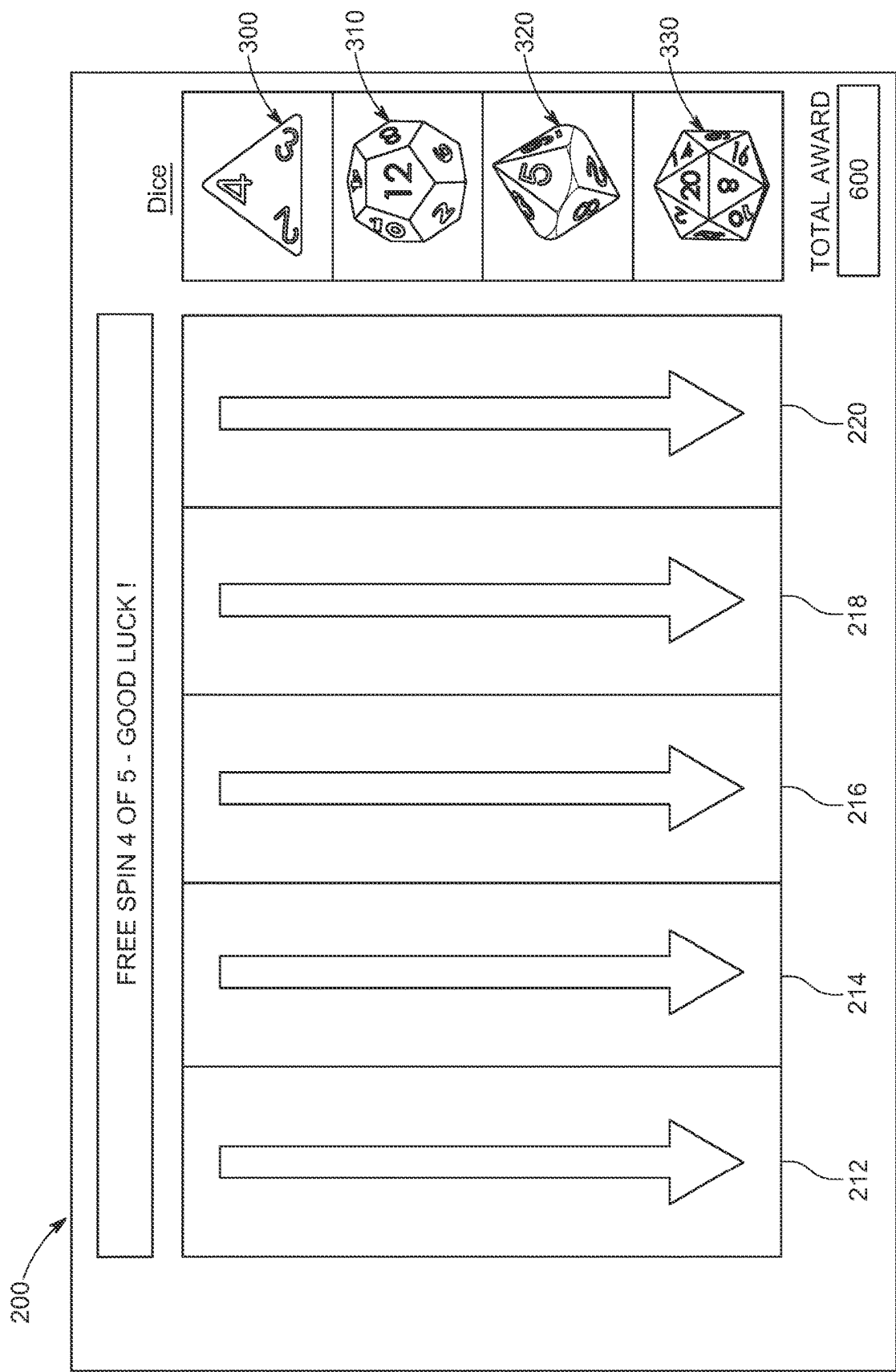


FIG. 2H

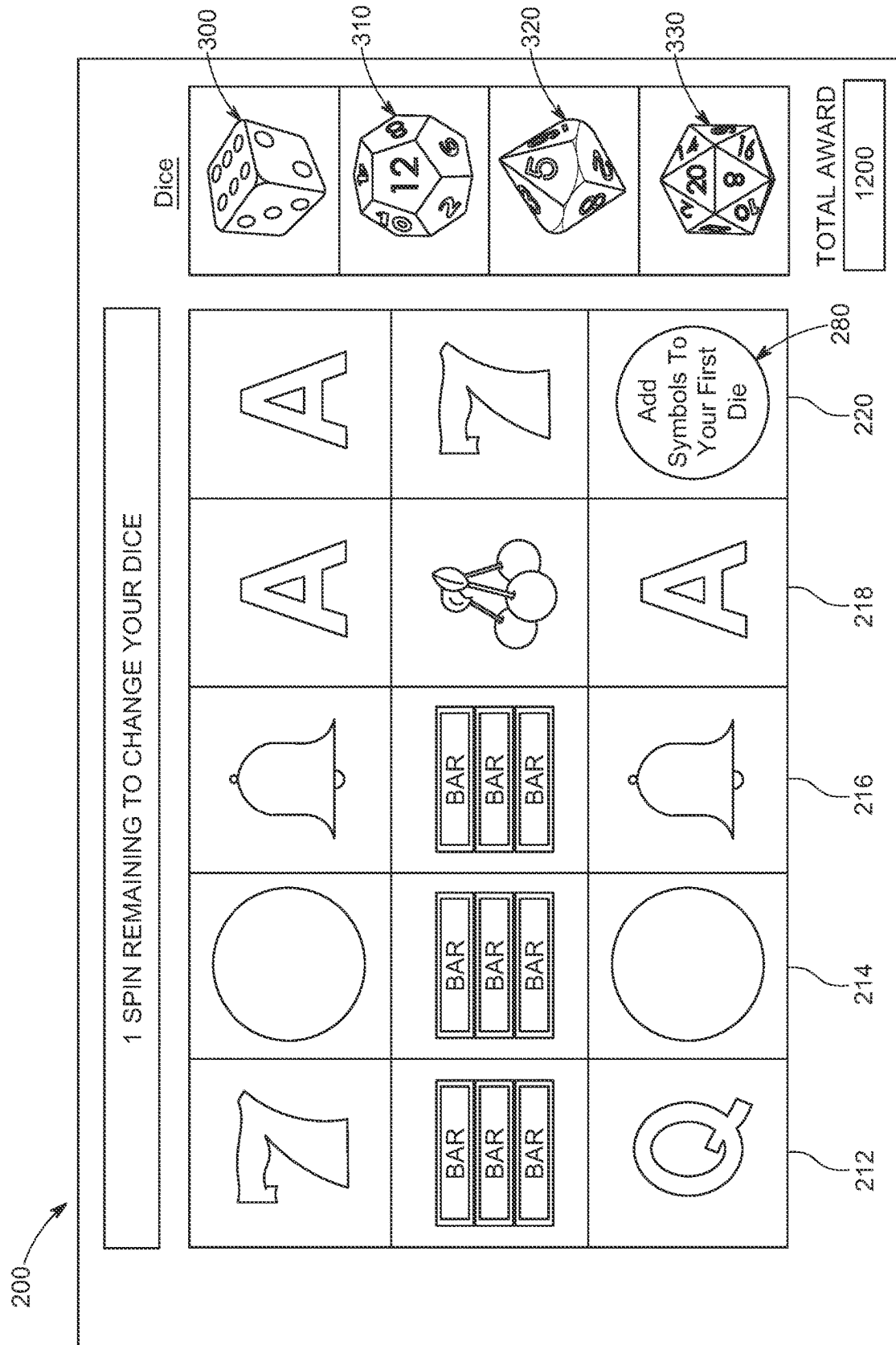


FIG. 21

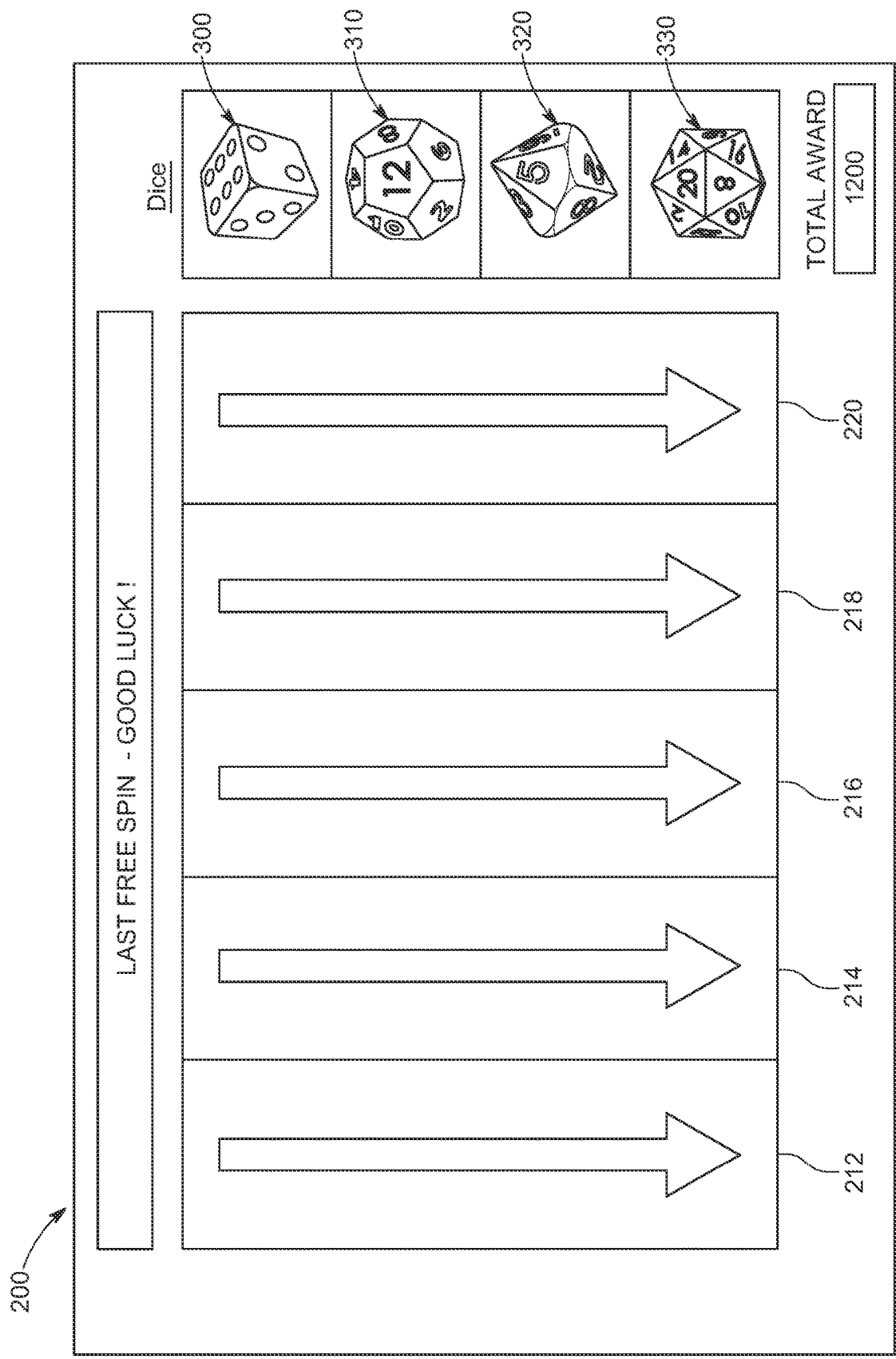


FIG. 2J

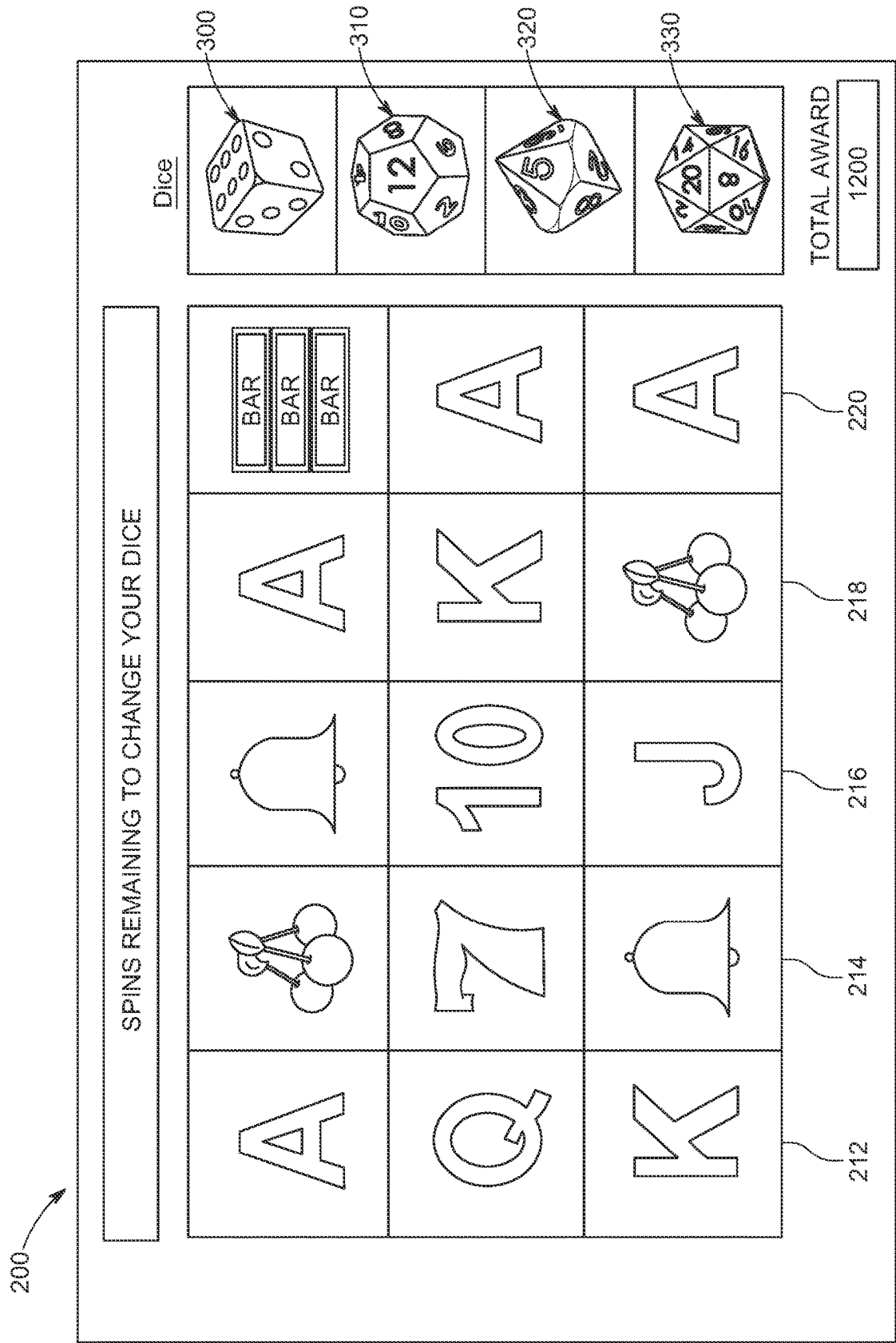


FIG. 2K

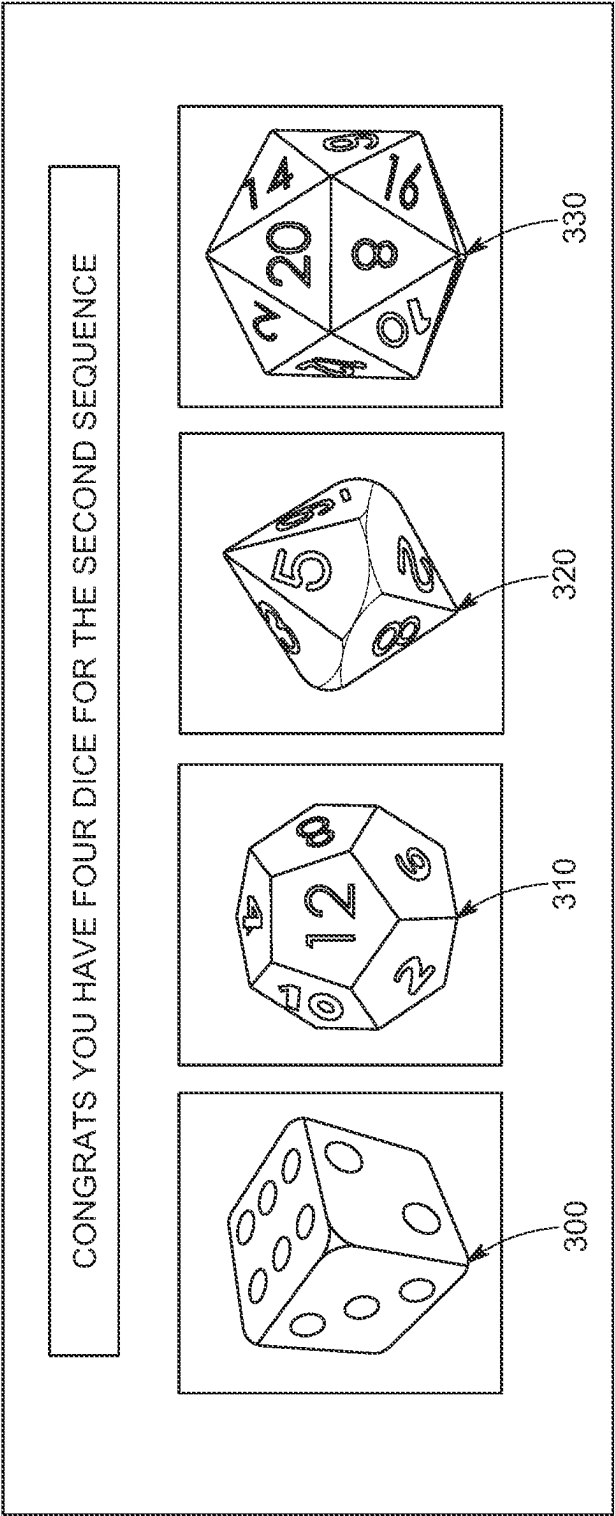


FIG. 3A

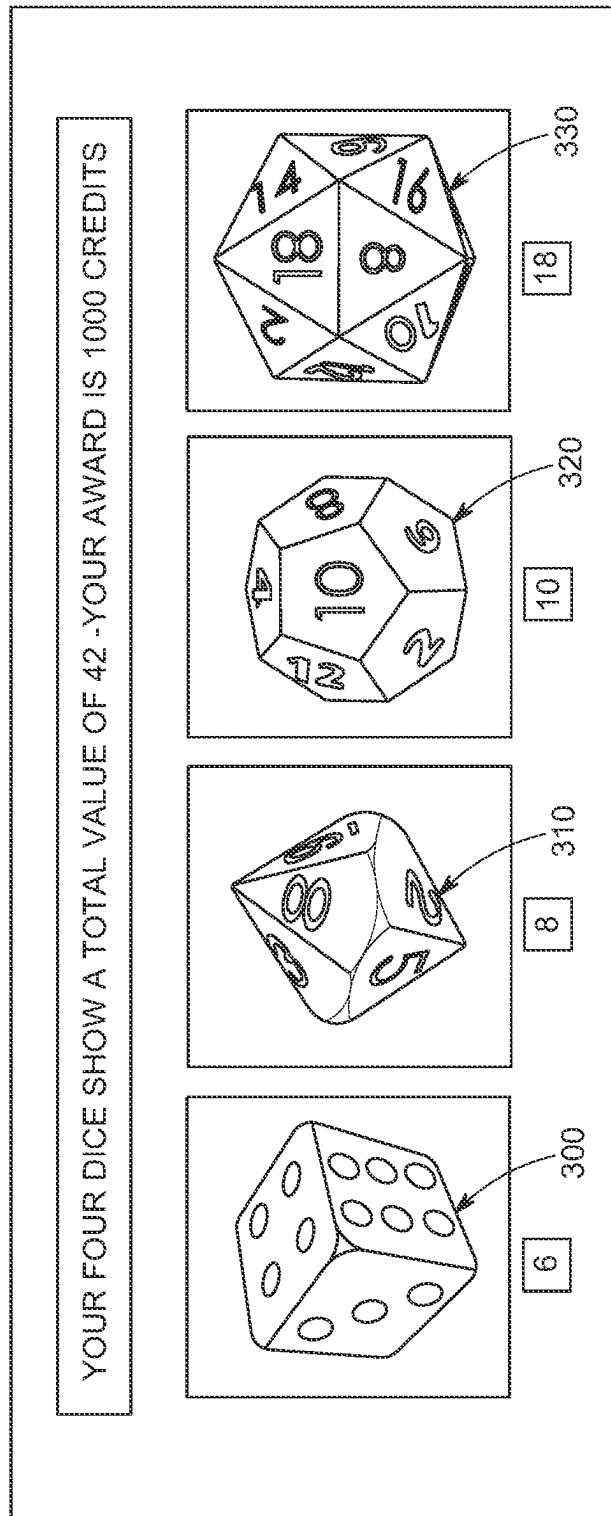


FIG. 3B

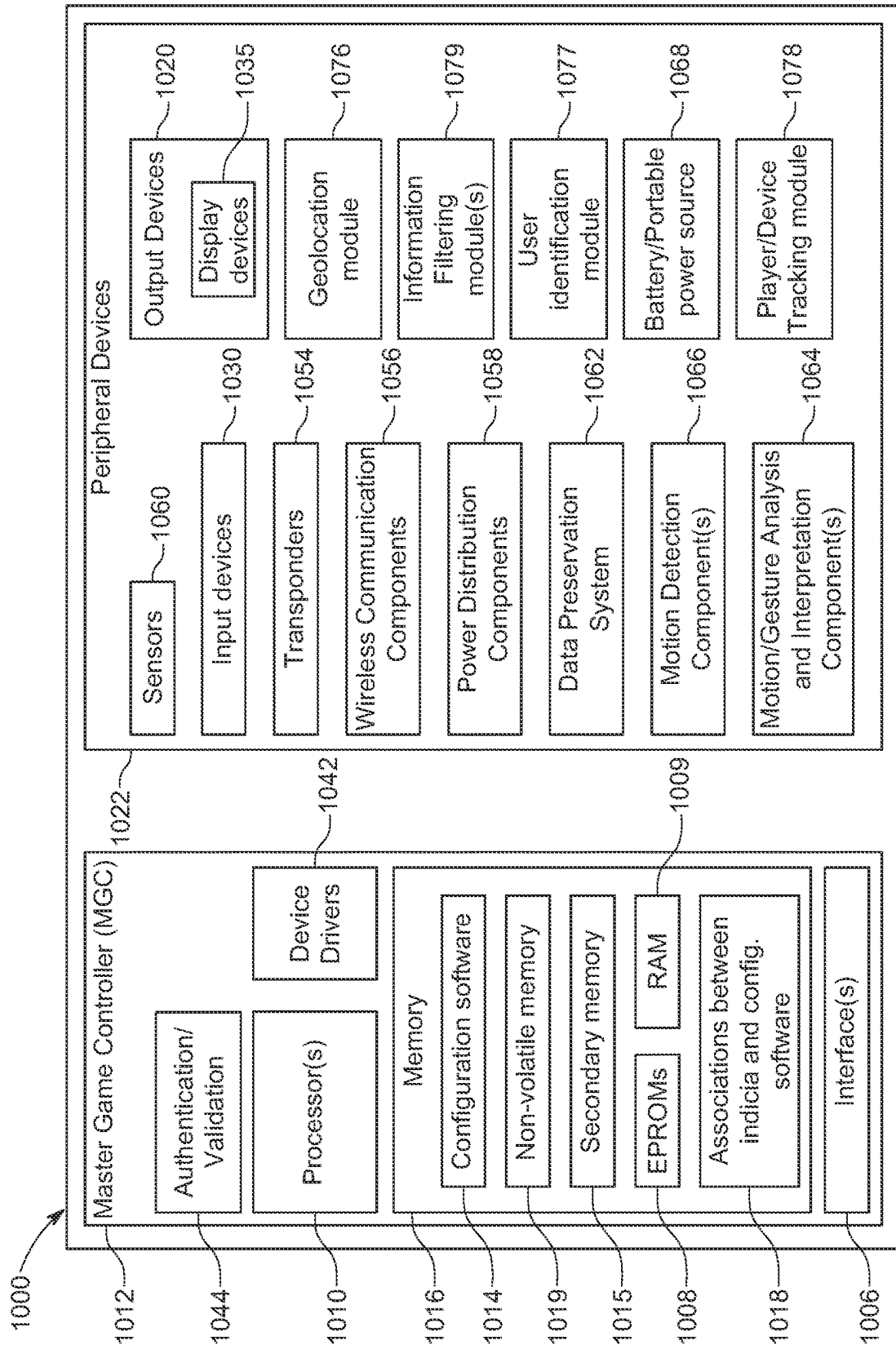


FIG. 4

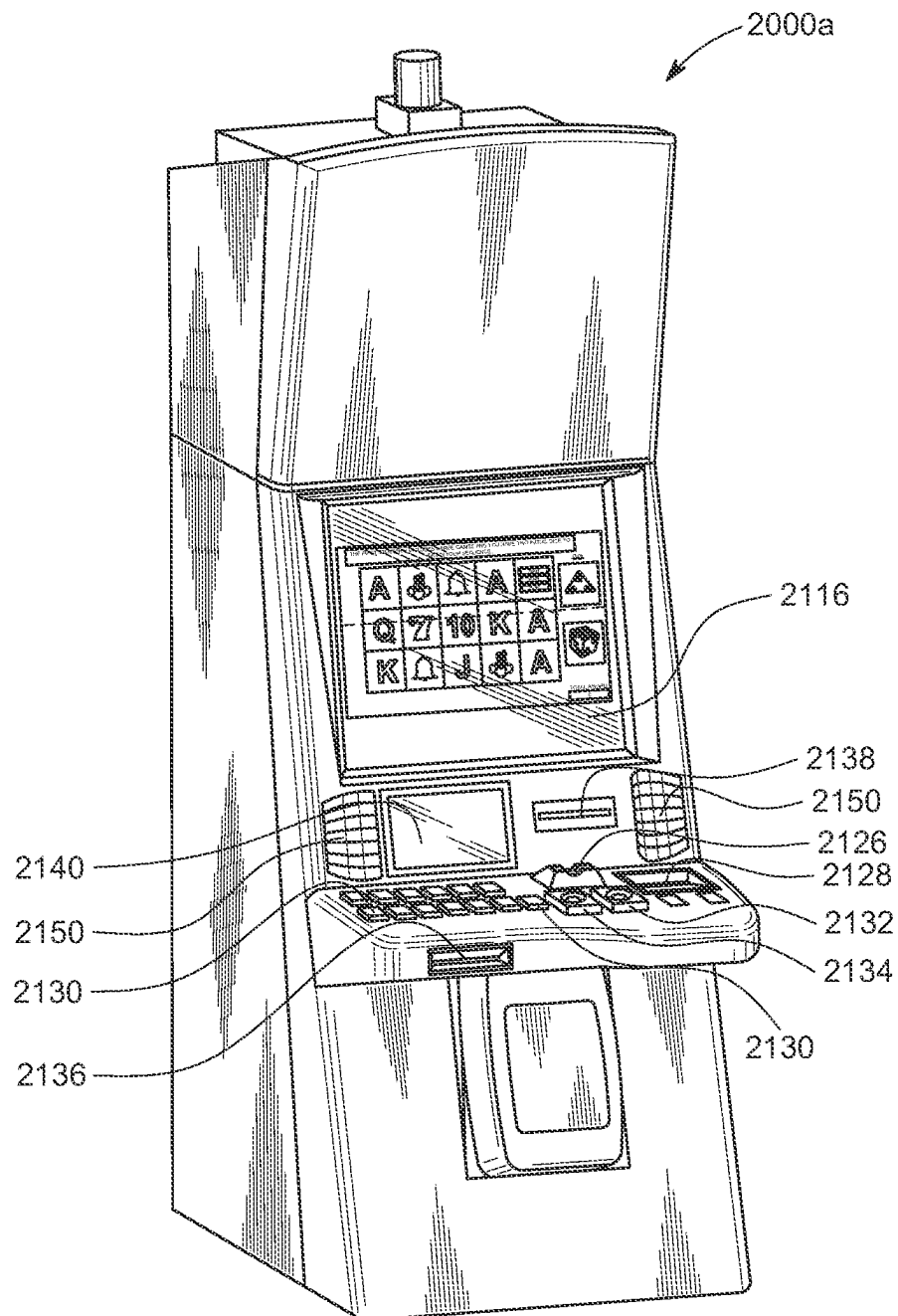


FIG. 5A

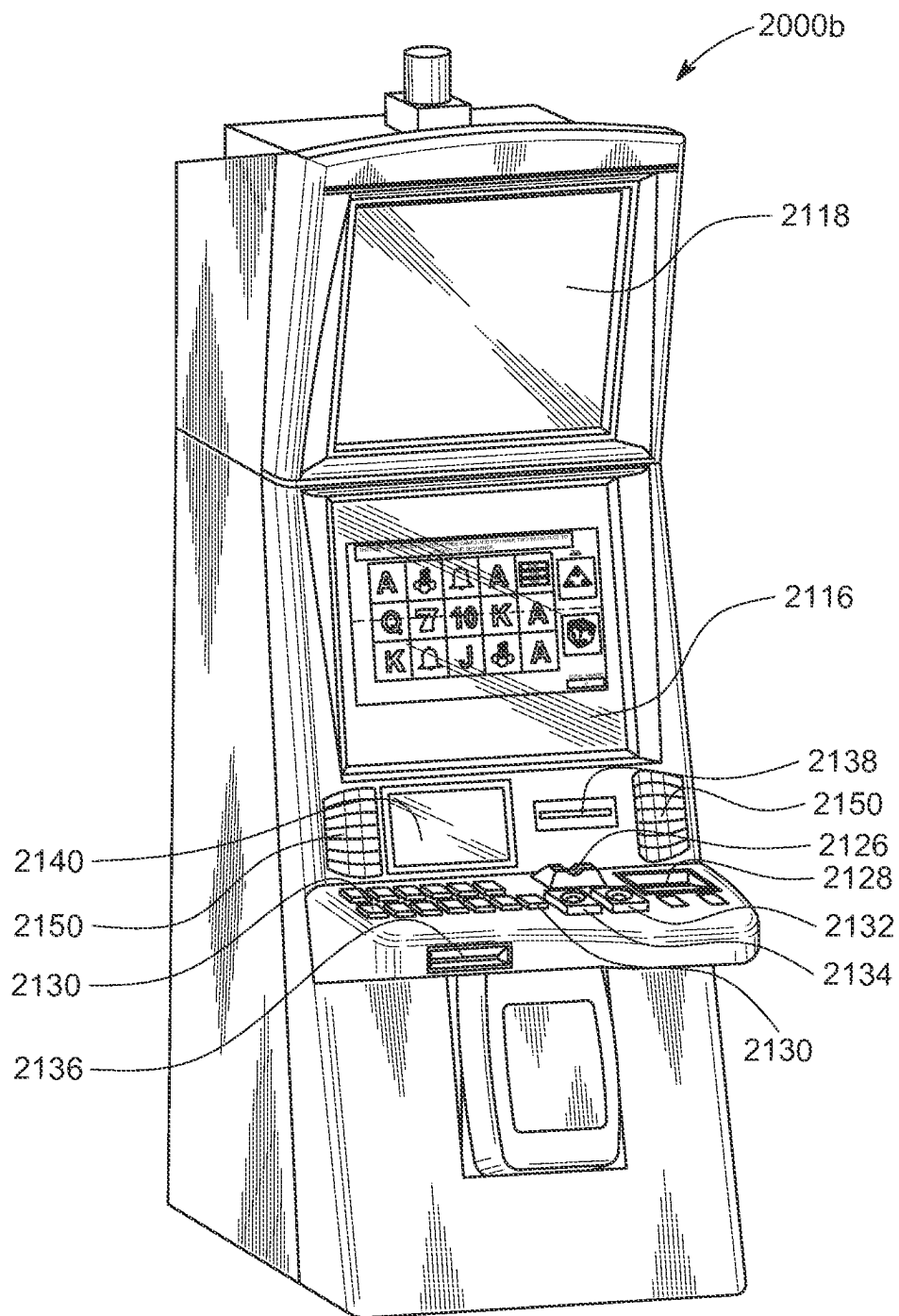
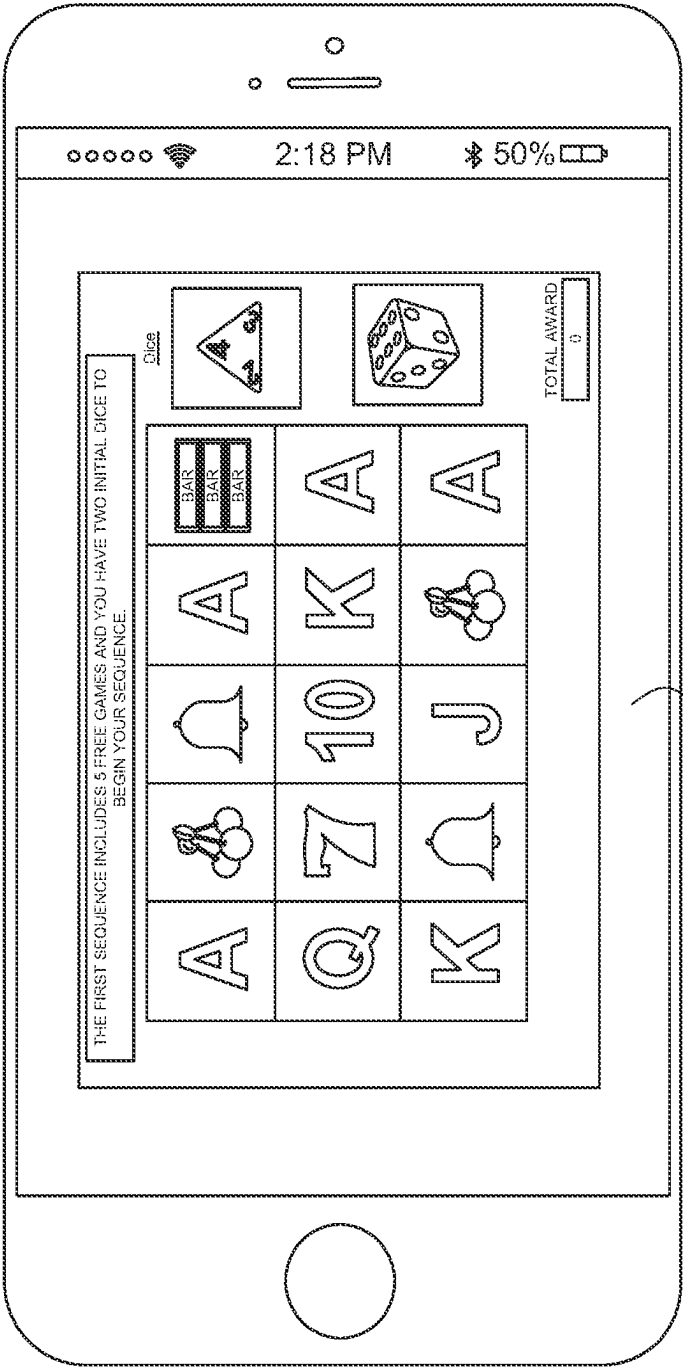


FIG. 5B

2000c



2116

FIG. 5C

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SYMBOL DISPLAY ACCUMULATION SEQUENCE AND A RELATED ACCUMULATED SYMBOL DISPLAY USE SEQUENCE

BACKGROUND

The present disclosure relates to a symbol display accumulation sequence and a related accumulated symbol display use sequence for gaming environments.

Gaming machines may require a player to place a wager to activate a play of the primary game. Gaming machines may provide one or more awards in a play of a primary game. Gaming machines may determine such awards based on a winning symbol or a winning symbol combination. Gaming machines may provide higher awards for winning symbols or winning symbol combinations that are less likely to occur in a play of a primary game. Gaming machines may provide a play of a secondary game responsive to an occurrence of a triggering event in a play of a primary game. Gaming machines may provide one or more additional awards in a play of a secondary game.

BRIEF SUMMARY

In various embodiments, the present disclosure relates to a gaming system including a processor and a memory device that stores a plurality of instructions that, when executed by the processor, cause the processor to: cause a display, by a display device, of a first sequence triggering event, cause a display, by the display device, of a first sequence including plays of a game, cause a display, by the display device, of a plurality of accumulated symbol displays that are generated during the first sequence, each accumulated symbol display including a quantity of different symbols, wherein two of the accumulated symbol displays includes different quantities of symbols, cause a display, by the display device, of the quantity of different symbols of one of the accumulated symbol displays changing during the first sequence, cause a display, by the display device, of a triggering event for a second sequence, and cause a display, by the display device, of the second sequence wherein one of the accumulated symbol displays remaining after the triggering event for the second sequence occurs is activated and displays one of the symbols on said accumulated symbol display.

In various other embodiments, the present disclosure relates to a gaming system including a processor and a memory device that stores a plurality of instructions that, when executed by the processor, cause the processor to: cause a display, by a display device, of a first sequence triggering event, cause a display, by the display device, of each of an initial quantity of accumulated symbol displays after the display of the first sequence triggering event, and for each said accumulated symbol display a quantity of symbols on said accumulated symbol display, the quantity being at least one, cause a display, by the display device, of a first sequence including plays of a game, cause a display, by the display device, of the accumulated symbol displays changing during the first sequence, and which includes a display, by the display device, of the quantity of different symbols on one of the accumulated symbol displays changing during the first sequence, and which includes a display, by the display device, of the quantity of the accumulated symbol displays changing during the first sequence, cause a display, by the display device, of a triggering event for a second sequence, and cause a display, by the display device, of the second sequence wherein each of the accumulated

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symbol displays remaining after the triggering event for the second sequence occurs is activated and displays one of the symbols on said accumulated symbol display.

In various other embodiments, the present disclosure relates to a gaming system including a processor and a memory device that stores a plurality of instructions that, when executed by the processor, cause the processor to: cause a display, by a display device, of a first sequence triggering event, cause a display, by the display device, of a first sequence including plays of a game, cause a display, by the display device, of a plurality of accumulated symbol displays that are displayed during the first sequence, each of the accumulated symbol displays including a quantity of different symbols, and two of the accumulated symbol displays including different quantities of symbols, cause a display, by the display device, of two of the accumulated symbol displays being formed into a combined accumulated symbol display during the first sequence, wherein the combined accumulated symbol display includes a sum of the quantities of symbols of the two accumulated symbol displays that are formed into the combined accumulated symbol display, cause a display, by the display device, of one of the accumulated symbol displays being split into two accumulated symbol displays during the first sequence, wherein the two accumulated symbol displays that are split from the one accumulated symbol display include a total quantity of symbols equal to the quantity of symbols of said one accumulated symbol display, cause a display, by the display device, of a triggering event for a second sequence, and cause a display, by the display device, of the second sequence wherein each of the accumulated symbol displays remaining after the triggering event for the second sequence occurs is activated and displays one of the symbols on said accumulated symbol display.

Additional features and advantages are described in, and will be apparent from, the following Detailed Description and the figures.

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

FIG. 1 is a flow chart of an example process for operating a gaming system of the present disclosure that provides a symbol display accumulation sequence and a related accumulated symbol display use sequence.

FIGS. 2A, 2B, 2C, 2D, 2E, 2F, 2G, 2H, 2I, 2J, and 2K are example screen shots displayed by a gaming system and illustrating parts of a plurality of plays of a game providing a symbol display accumulation sequence in accordance with one example embodiment of the present disclosure.

FIGS. 3A and 3B are example screen shots displayed the gaming system and illustrating parts of an accumulated symbol display use sequence in accordance with one example embodiment of the present disclosure.

FIG. 4 is a schematic block diagram of one example embodiment of an electronic configuration of a gaming system of the present disclosure.

FIGS. 5A and 5B are perspective views of example alternative embodiments of the gaming system of the present disclosure.

FIG. 5C is a front view of an example personal gaming device of the gaming system of the present disclosure.

DETAILED DESCRIPTION

In various embodiments, the present disclosure relates to gaming systems and methods that provide a symbol display

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accumulation sequence and a related accumulated symbol display use sequence. The symbol display accumulation sequence is sometimes referred to herein as an accumulation sequence or as a first sequence for brevity, and the related accumulated symbol display use sequence is sometimes referred to herein as a use sequence or as a second sequence for brevity. The accumulation sequence generally includes an accumulation of symbol displays during one or more plays of one or more games during such accumulation sequence. The use sequence generally includes the use of the accumulated symbol displays.

In various embodiments, responsive to a triggering event, the gaming system determines: (1) an initial quantity of accumulated symbol displays to provide for the first sequence; (2) for each such symbol display, how many symbols to provide on such symbol display; and/or (3) for each such symbol on each such symbol display, which symbols of a plurality of different symbols to provide on such symbol display. The gaming system then provides and displays indications of such initial symbol displays for the first sequence at the start of the first sequence. In various embodiments, the gaming system randomly determines: (1) the initial quantity of accumulated symbol displays to provide for the first sequence; (2) for each such symbol display, how many symbols to provide on such symbol display; and/or (3) for each such symbol on each such symbol display, which symbols of a plurality of different symbols to provide on such symbol display, are predetermined (such as in a static manner, or based on another factor such as a wager made for a triggering play of the game or a triggering event). These determinations can thus be made in any suitable manner in accordance with the present disclosure.

In various embodiments, the gaming system then provides and displays the first sequence wherein one or more events of the first sequence can cause one or more changes to: (1) the quantity of accumulated symbol displays; (2) for each accumulated symbol display, how many symbols are on such accumulated symbol display; and/or (3) for each symbol on each such accumulated symbol display, a change to such symbol. The gaming system then provides and displays indications of the final accumulated symbol displays at the end of the first sequence prior to the start of the second sequence. In various embodiments, during the first sequence, the gaming system can perform one or more or each of the following functions: (1) increasing the quantity of accumulated symbol displays; (2) decreasing the quantity of accumulated symbol displays; (3) increasing the quantity of symbols on one or more or each of the accumulated symbol displays; (4) decreasing the quantity of symbols on one or more or each of the accumulated symbol displays; and (5) changing one or more or each of the symbols on one or more or each of the accumulated symbol displays. Thus, in various such embodiments, the gaming system can modify the symbol displays in the first sequence for use in the second sequence. The first sequence thus in part determines the available outcomes of the second sequence.

In various embodiments of the present disclosure, for each play of each game of the first sequence, the gaming system: (1) randomly determines a plurality of symbols; (2) displays the randomly determined symbols at a plurality of

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symbol display positions; (3) determines any awards based on those displayed symbols; (4) displays any determined awards; (5) accumulates any additional symbol displays and/or discards any lost accumulated symbol displays; and (6) changes any of the accumulated symbol displays.

In various embodiments, the gaming system then provides and displays the second sequence wherein one or more events of the second sequence uses one or more of the final accumulated symbol displays with the final symbols on such accumulated symbol displays. In various embodiments, the gaming system uses all of the final accumulated symbol displays for the second sequence. In various embodiments, the gaming system uses less than all of the final accumulated symbol displays for the second sequence.

In various embodiments of the present disclosure, responsive to an occurrence of a second sequence triggering event, the gaming system provides the second sequence. For example, in various embodiments, the second sequence triggering event includes the accumulation of a triggering quantity of symbol displays during the first sequence. In various embodiments, the triggering quantity of accumulated symbol displays during the first sequence is a static quantity such as six accumulated symbol displays. It should be appreciated that the second sequence triggering event can occur based on one or more other conditions in accordance with the present disclosure.

In various embodiments of the present disclosure, for the second sequence, the gaming system employs the previously accumulated symbol displays generated and displayed during the first sequence for the subsequent second sequence. In various embodiments, for the second sequence, the gaming system at least partly determines any awards based on the accumulated symbol displays from the first sequence, based on a suitable payable for the accumulated symbol displays, and for each accumulated symbol display, a randomly determined one of the symbols of that accumulated symbol display.

Such a configuration of enabling a player to accumulate symbol displays that are then used for one or more functions including: (1) for triggering the second sequence; and (2) for determining awards in the second sequence, increases the player's anticipation as more symbol displays are accumulated for the play(s) of the game of the first sequence.

The present disclosure employs dice as one example embodiment of the accumulated symbol displays of the present disclosure to explain the present disclosure. However, the present disclosure contemplates that any suitable symbol displays can be employed in accordance with the present disclosure, and thus the use of dice as example accumulated symbol displays are not meant to limit the present disclosure.

While certain embodiments described herein are directed to a plurality of plays of a secondary game (such as free games) that are part of the first sequence, it should be appreciated that such embodiments may additionally or alternatively be employed in association with primary games, such as a plurality of plays of a primary games that are part of the first sequence.

Additionally, while the player's credit balance, the player's wager, and any awards are displayed amounts of monetary credits or currency in certain of the example embodiments described herein, one or more of such player's credit balance, such player's wager, and any awards provided to such a player may be for non-monetary credits, promotional credits, and/or player tracking points or credits.

FIG. 1 is a flowchart of an example method 100 of operating a gaming system of the present disclosure. In

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various embodiments, the method is represented by a set of instructions stored in one or more memories and executed by one or more processors. Although the method is described with reference to the flowchart shown in FIG. 1, many other processes of performing the acts associated with this illustrated method may be employed. For example, the order of certain of the illustrated blocks may be changed, certain of the illustrated blocks may be optional, or certain of the illustrated blocks may not be employed.

In this example embodiment, responsive to a first triggering event for a symbol display accumulation sequence, the gaming system determines and displays an initial quantity of accumulated symbol displays, as indicated by block 102.

The gaming system displays a symbol display accumulation sequence in which the gaming system can change the quantity of accumulated symbol displays and change the symbols on each of the accumulated symbol displays, as indicated in block 104. In various embodiments, during the symbol display accumulation sequence, the gaming system: (1) randomly determines a plurality of symbols; (2) displays the randomly determined symbols at a plurality of symbol display positions; (3) determines any awards based on those displayed symbols; (4) displays any determined awards; (5) accumulates any additional symbol displays and/or discards any lost symbol displays; and (6) changes any of the accumulated symbol displays. In various embodiments, the symbol display accumulation sequence includes one or more plays of a game (such as a primary wagering game), and for each play of the game, the gaming system accumulates any additional symbol displays that are generated during that play of the game and makes any changes to the accumulated symbol displays. The gaming system causes any accumulated symbol displays in the symbol display accumulation sequence to continue to be displayed for each subsequent play of the game of the accumulation sequence until a second sequence triggering event occurs.

Responsive to the second sequence triggering event occurring, the gaming system provides the accumulated symbol display use sequence, as indicated by block 106. The accumulated symbol display use sequence includes randomly determining for each accumulated symbol display, a symbol on that accumulated symbol display, displaying the randomly determined symbols on the accumulated symbol displays, determining an award based on those displayed symbols, and displaying the determined award, as also indicated by block 106. The gaming system also provides the determined award to the player, as also indicated by block 106.

It should be appreciated that enabling a player to play the first sequence to accumulate symbol displays (which are then used in the second sequence) increases the player's anticipation as more symbol displays are accumulated during the first sequence.

FIGS. 2A, 2B, 2C, 2D, 2E, 2F, 2G, 2H, 2I, 2J, and 2K illustrate parts of a plurality of plays of a game of an example first sequence wherein the symbol displays are accumulated and changed in accordance with one example embodiment of the gaming system of the present disclosure. In this example, the display device 200 of the gaming system displays free plays of a slot game including five reels, 212, 214, 216, 218, and 220 and a plurality of different symbols (not labelled) on each reel. It should also be appreciated that any suitable quantity of reels and any suitable symbols can be employed in accordance with the present disclosure.

FIG. 2A illustrates the beginning of the first sequence before the first game play of the first sequence has occurred. The gaming system has determined the initial quantity of

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accumulated symbol displays that have been determined for the first sequence. In this example, two symbol displays (four-sided die 300 and six-sided die 310) are accumulated at this point in time, and the gaming system is ready to provide the first free play of the first sequence. FIG. 2A further illustrates the respective symbols on those accumulated symbol displays 300 and 310.

FIGS. 2B and 2C illustrate the gaming system spinning the reels 212, 214, 216, 218, and 220 and stopping the reels to display the randomly determined symbols on the reels at a plurality of symbol display positions for this first free play of the game of the first sequence. The gaming system determines any awards based on those displayed symbols, and as illustrated in FIG. 2C displays any determined awards. In this example, the gaming system determines the award of 200 credits based on the three Q symbols. As further shown in FIG. 2C, the gaming system accumulates an additional symbol display based on the occurrence of the displayed designated symbol 250 from that play of the game of the first sequence. The gaming system causes this additional symbol display (twelve sided die 320) to be displayed as shown in FIG. 2C. The gaming system continues to display the accumulated symbol displays 300, 310, and 320 for the next play of the game of the first sequence as illustrated in FIGS. 2D and 2E. The gaming system also determines at this point if a second sequence triggering event has occurred. In this example, the second game triggering event is no further free spins remaining, and the gaming system determines that the second sequence triggering event has not occurred at this point in this example.

The gaming system causes the second free play of the game. FIGS. 2D and 2E illustrate parts of the second free play of the game including the gaming system spinning the reels and stopping the reels 212, 214, 216, 218, and 220 to display the randomly determined symbols at a plurality of symbol display positions for this second free play of the game of the first sequence. The gaming system determines any awards based on those displayed symbols, and as illustrated in FIG. 2E displays any determined awards. In this example, the gaming system determines the award of 100 credits based on the three J symbols. As further shown in FIG. 2E, the gaming system does not accumulate any further symbol displays from this second free play of the game of the first sequence, but doubles the quantity of symbols on the second symbol display (die 310) based on the occurrence of the Double Your 2nd Die triggering symbol 260. In this example, the gaming system doubles the quantity of symbols on the second accumulated symbol display 310 from six to twelve. The gaming system continues to display the accumulated symbol displays 300, 310, and 320 for the next play of the game of the first sequence as illustrated in FIGS. 2F and 2G. The gaming system also determines at this point if a second sequence triggering event has occurred. In this example, the second game triggering event is no further free spins remaining, and the gaming system determines that the second sequence triggering event has not occurred at this point in this example.

The gaming system causes the third free play of the game. FIGS. 2F and 2G illustrate parts of third free play of the game including the gaming system spinning the reels and stopping the reels 212, 214, 216, 218, and 220 to display the randomly determined symbols at a plurality of symbol display positions for this third free play of the game of the first sequence. The gaming system determines any awards based on those displayed symbols, and as illustrated in FIG. 2G displays any determined awards. In this example, the gaming system determines the award of 300 credits based on

the three K symbols. As further shown in FIG. 2G, the gaming system accumulates a further symbol display (twenty-sided die 330) from this third play of the game of the first sequence because the symbol 270 is generated and displayed. The gaming system adds this additional accumulated symbol display 330. The gaming system continues to display the accumulated symbol displays 300, 310, 320, and 330 for the next play of the game of the first sequence as illustrated in FIGS. 2H and 2I. The gaming system also determines at this point if a second sequence triggering event has occurred. In this example, the second game triggering event is no further free spins remaining, and the gaming system determines that the second sequence triggering event has not occurred at this point in this example.

The gaming system causes the fourth free play of the game. FIGS. 2H and 2I illustrate parts of fourth free play of the game including the gaming system spinning the reels and stopping the reels 212, 214, 216, 218, and 220 to display the randomly determined symbols at a plurality of symbol display positions for this fourth free play of the game of the first sequence. The gaming system determines any awards based on those displayed symbols, and as illustrated in FIG. 2I displays any determined awards. In this example, the gaming system determines the award of 600 credits based on the three BAR-BAR-BAR symbols. As further shown in FIG. 2I, the gaming system does not accumulate any further symbol displays from this fourth free play of the game of the first sequence but adds two symbols to the first accumulated symbol display (four-side die 300) based on the occurrence of the Add Symbols to Your First Die triggering symbol 280. In this example, the gaming system add two symbols and thus the four-sided die 300 becomes a six-sided die 300. The gaming system continues to display the accumulated symbol displays 300, 310, 320, and 330 for the next play of the game of the first sequence as illustrated in FIGS. 2J and 2K. The gaming system also determines at this point if a second sequence triggering event has occurred. In this example, the second game triggering event is no further free spins remaining, and the gaming system determines that the second sequence triggering event has not occurred at this point in this example.

The gaming system causes the fifth free play of the game. FIGS. 2J and 2K illustrate parts of the fifth free play of the game including the gaming system spinning the reels and stopping the reels 212, 214, 216, 218, and 220 to display the randomly determined symbols at a plurality of symbol display positions for this fifth free play of the game of the first sequence. The gaming system determines any awards based on those displayed symbols. In this example, no awards occur based on the displayed symbols. As further shown in FIG. 2I, the gaming system does not accumulate any further symbol displays from this fifth free play of the game of the first sequence and does not change any of the accumulated symbol displays 300, 310, 320, and 330. The gaming system also determines at this point if a second sequence triggering event has occurred. In this example, as mentioned above, the second triggering event is no free plays remaining and the gaming system determines that the second sequence triggering event has occurred.

An example second sequence is partly illustrated in FIGS. 3A and 3B. In this example, the gaming system uses the accumulated symbol displays 300, 310, 320, and 330 from the first sequence for the second sequence. FIG. 3A illustrates the accumulated symbol displays 300, 310, 320, and 330 for the second sequence. The second sequence in this example includes the gaming system randomly determining for each of the accumulated symbol displays 300, 310, 320,

and 330 one of the symbols on that accumulated symbol display and displaying that randomly determined symbol on that accumulated symbol display. This is shown FIG. 3B. In this example, the gaming system also displays separate indications of each of the randomly symbols of the accumulated symbol displays 300, 310, 320, and 330 below the respective accumulated symbol display. In this example, the gaming system determines the award based on a sum of the values of the randomly determined displayed symbols on the accumulated symbol displays 300, 310, 320, and 330. The gaming system also provides the determined award to the player. After providing the determined award, the gaming system ends the second sequence, the gaming system returns to the primary game.

It should be appreciated from the above that the present disclosure employs the accumulated symbol displays (such as the dice) to display the future opportunities in the second sequence.

It should be appreciated from the above that the gaming system enables the player to obtain additional accumulated symbol displays and upgrade (or otherwise change) the accumulated symbol displays (such as dice) to increase the chances of winning better awards.

It should be appreciated from the above that the gaming system can provide an initial quantity of accumulated symbol displays (such as dice), but that in other embodiments, the gaming system can provide no initial accumulated symbol displays.

It should be appreciated from the above that the types, shapes, sizes, and other characteristics of the accumulated symbol displays (such as dice) can vary in accordance with the present disclosure. For example, the dice can have a variety of different shapes and sides such a four, six, ten, twelve, and twenty sided dice.

It should be appreciated from the above, and as further provided below, that the quantity of available symbols on the symbol displays that can be accumulated can be employed in combination for determining awards for the player.

It should further be appreciated that the present disclosure also contemplates that the accumulated symbol displays can have other factors or characteristics (such as color) that can be associated with different functions such as different awards. For example, certain color accumulated symbol displays (such as dice) could be associated with progressive awards or multipliers. In various embodiments, the color, size, state, or animation of the accumulated symbol display (such as the die) could indicate the meaning of the accumulated symbol display (such as the die). For example, blue dice could be used in the award determination phase of the second sequence to determine a payable win, and yellow dice could be used in the award determination phase of the second sequence to determine a progressive award. Thus, in various embodiments, the color, size, state, and animation of the accumulated symbol displays (such as the dice) could represent the type of evaluation, the payable used for the evaluation, or the type of award (such a credit award, a multiplier, or points). In other embodiments, the color, size, state, and animation of the accumulated symbol displays (such as the dice) do not represent any functions, features, or awards.

In various embodiments (such as the above described embodiments), the gaming system selects which accumulated symbol displays to change. In various other embodiments, the gaming system enables the player to select which accumulated symbol displays to change (when an accumulated symbol display change event occurs). In various embodiments, the gaming system enables the player to try

for different awards by selecting the changes to the accumulated symbol displays (such as dice) during the first sequence. In various alternative embodiments, the gaming system enables the changes to the symbol displays such as by combining multiple accumulated symbol displays (such as combining dice) or dividing accumulated symbol displays into multiple accumulated symbol displays (such as breaking a die into multiple dice). In various such embodiments, the gaming system controls such combinations and dividing actions. In various such embodiments, the gaming system enables the player to control such combinations and dividing actions. In various embodiments, the gaming system or the player can have one or more opportunities to take such actions.

In the above described example embodiment, the accumulated symbol display changing events occur based on triggering symbols that occur during the free plays of the game. In various other embodiments, the accumulated symbol display changing events can occur based on other triggering events. For example, the accumulated symbol display changing events can occur on a regular basis (such as an event for every free play, every other free play, or every fifth free play). In various other embodiments, the accumulated symbol display changing events can occur based on certain winning symbol combinations occurring, certain winning amounts occurring, or some random event occurring (such as a lucky coin event from a host system).

In other embodiments, the gaming system performs an evaluation to determine a quantity of symbol displays to add or changes to an accumulated symbol display (such as side added to one or more of the accumulated dice). For example, a 3-cherry winning symbol combination during the first sequence can cause the gaming system to add one symbol or side to an existing die and a 4-cherry winning symbol combination during the first sequence can cause the gaming system to add two symbols or sides to an existing die. In alternative embodiments, after a triggering symbol occurs during the first sequence to cause a change to one or more of the accumulated symbol displays, the quantity of symbols (such as sides of a die) added to one or more of the accumulated symbol displays (such as a die) could be determined by the gaming system in a random manner or in any other suitable manner.

In various embodiments, in the second sequence, the accumulated symbol displays can be activated in any suitable manner. For example, the gaming system can automatically activate the accumulated symbol displays or enable the player to activate the accumulated symbol displays (such as by causing the rolling of the dice such as by touching the display of the dice on a touch screen, activating a button, swiping on the touch screen, using a hand gesture, using a voice command, or via a separate player device such as a player mobile device). In various embodiments, the accumulated symbol display movement (such as the dice movement) is a graphical representation on the screen of the gaming system that mimics the physical behaviors of the accumulated symbol displays (such as the rolled accumulated dice). The symbol display (such as the dice) movement be provided via a physics engine within gaming system, or a precalculated animation to simulate realistic symbol display (such as dice) movement or predict an outcome matching a payable.

In various embodiments, in the second sequence, the accumulated symbol displays can be activated in any suitable order or can be activated simultaneously. For example, in one embodiment, the gaming system can enable the player to roll the dice individually while in another embodiment the

gaming system can enable the player to roll all dice at once. In certain embodiments, the order is randomly selected by the gaming system. In certain embodiments, the order is predetermined.

In certain embodiments, the gaming system can enable the player to choose which dice to select and try for a certain outcome. For example, some dice (blue) can provide certain outcomes and other dice (green) can produce other outcomes; and the player may choose to roll the green dice because those outcomes are more appealing to the player. In certain such embodiments, the gaming system enables the player may choose to roll multiple "colors" of dice to hedge their bets.

In various embodiments, the paytables associated with the accumulated symbol displays can be based on the accumulated symbol displays and the symbols on the actual accumulated symbol displays or can be independent of the actual accumulated symbol displays. The following are examples of such paytables using example accumulated dice.

In a first example, the accumulated dice are two twenty-sided dice and three six-sided dice. The additive value of these dice range from a low of 5 to a high of 58.

The following payable can be used for this range for this first example wherein the range matches the possible outcomes of the actual accumulated dice.

Range	Payback
5-10	\$5.00
11-20	\$6.00
21-50	\$15.00
51-57	\$40.00
58	\$100.00

Alternatively, the gaming system could use the following payable because more and different dice could have been accumulated in the first sequence and thus the range does not only match the possible outcomes of the actual accumulated dice.

Range	Payback
1-10	\$5.00
11-50	\$10.00
51-100	\$100.00
101-250	\$500.00
251	\$10,000.00

Alternatively, the gaming system could use the following payable with a progressive award because more and different dice could have been accumulated in the first sequence and thus the range does not match the possible outcomes of the actual accumulated dice.

Range	Payback
1-10	\$5.00
11-50	\$10.00
51-100	\$100.00
101-250	\$500.00
251	PROGRESSIVE

While the second and third example paytables show that an increasing total value resulting from the randomly determined symbols from the accumulated dice yield a higher award, the present disclosure contemplates that other paytables are possible. For example, if a player has one twenty-

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sided dice in one second sequence and the same player as three six-sided dice in another second sequence, the gaming system can be configured to guarantee values greater than or equal to 3 and have the distribution of values centered around 10.5. This is because the expected value of one six-sided die is $(1+2+3+4+5+6)/6=3.5$, $3.5*3=10.5$. Therefore, there is a distribution around 10.5 for the expected values of 3 six-sided dice. However, with one twenty-sided dice, each value has the exact same odds (1 in 20). This enables a game designer to create paytables where the player tries to hit values in a certain range instead of the larger value and therefore would cause the player to choose different dice to roll or make difference decisions during the accumulation sequence.

In other embodiments, the gaming system enables the player to roll certain of the accumulated dice that the gaming system evaluates using a first payable and then roll certain of the accumulated dice that the gaming system evaluates against a different second payable. For example, the player can roll a first set of accumulated dice to determine a payable win and then a second set of accumulated dice to determine the progressive win. It should be appreciated that the different paytables can provide credits, points, free spins, bonus spins, multipliers, or progressives awards.

In various embodiment, the values of the symbols of the accumulated dice in the second sequence include a multiplication of the evaluation value. For example, rolling a 10, 20, and 5 on three accumulated dice can result in the value of $10 \times 20 \times 5 = 1,000$ when applied to a payable or otherwise determining an award. In another example, if the player rolls two red dice for values 10 and 20 and a blue die for the value of 5, the red dice can be additive, and the blue die can be multiplicative resulting in an outcome of $(10+20) \times 5 = 150$ for being applied to a payable or otherwise determining an award.

In various embodiments, the order of the evaluation can vary. For example, if the accumulated dice values are 1, 2, and 5, the evaluation can be $1 \times 2 + 5$, or can be $2 \times 5 + 1$ which is better. In various embodiments, the order of evaluation is predetermined. In various embodiments, the order of the evaluation is randomly determined. In various embodiments, the order of evaluation used is the order that results in the highest award for the player. In various other embodiments, the order can be based on one or more characteristics of the dice (such as red, then yellow then blue). In another example, the order can be $\text{red} \times \text{yellow} + \text{blue}$. In various other embodiments, the gaming system enables the player to partially or fully pick the order of evaluation. For example, the gaming system can enable the player to pick the first die to use in the evaluation. In various other embodiments, the gaming system can enable the player to win upgraded operators. For example, the gaming system can enable one or more of the multipliers to be won to replace one or more of the addition operators. For example, the player can start with a $2+5+6$ dice value evaluation, then upgrade to a $2 \times 5+6$ dice value evaluation, and then upgrade to a $2 \times 5 \times 6$ dice value evaluation.

In various embodiments, the free plays of the game of the first sequence provide monetary awards and then the accumulated dice roll of the second sequence results in a multiplier for that award.

In various embodiments, the value of the final accumulated dice roll, whether additive, multiplicative, or any combination, is the final award. For example, if the accumulated dice evaluation results in the value $10 \times 20 \times 5 = 1,000$, the player is awarded \$1,000 for the second sequence.

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In various embodiment, after rolling the accumulated dice, the player may touch or select one or more dice that expose an additional value. This value could be monetary, a multiplier, or points. The value or multiplier is applied to the accumulated dice evaluation value.

In various embodiments where the players get to choose the quantity of accumulated dice to roll for the evaluation, the quantity of accumulated dice could be determined by one or more events in the free games, a triggering event from the primary game, or another random event.

In various embodiments, the player has multiple rolls to determine the outcome of the second sequence. Each roll selects a subset on which to further evaluate. For example, the player first rolls one accumulated die to determine which progressive level to attempt to win (e.g., the mini, the major, or the grand progressive award); and a second roll determines if the player wins the level chosen from the first roll.

In various embodiments, one or more awards provided in association with the games disclosed herein include one or more of: a quantity of monetary credits, a quantity of non-monetary credits, a quantity of promotional credits, a quantity of player tracking points, a progressive award, a modifier, such as a multiplier, a quantity of free plays of one or more games, a quantity of plays of one or more secondary or bonus games, a multiplier of a quantity of free plays of a game, one or more lottery based awards, such as lottery or drawing tickets, a wager match for one or more plays of one or more games, an increase in the average expected payback percentage for one or more plays of one or more games, one or more comps, such as a free dinner, a free night's stay at a hotel, a high value product such as a free car, or a low value product, one or more bonus credits usable for online play, a lump sum of player tracking points or credits, a multiplier for player tracking points or credits, an increase in a membership or player tracking level, one or more coupons or promotions usable within and/or outside of the gaming establishment (e.g., a 20% off coupon for use at a convenience store), virtual goods associated with the gaming system, virtual goods not associated with the gaming system, an access code usable to unlock content on an internet.

In various embodiment, any suitable determination disclosed herein can be predetermined, randomly determined, randomly determined based on one or more weighted percentages, determined based on a generated symbol or symbol combination, determined independent of a generated symbol or symbol combination, determined based on a random determination by the central controller, determined independent of a random determination by the central controller, determined based on a random determination at the gaming system, determined independent of a random determination at the gaming system, determined based on at least one play of at least one game, determined independent of at least one play of at least one game, determined based on a player's selection, determined independent of a player's selection, determined based on one or more side wagers placed, determined independent of one or more side wagers placed, determined based on the player's primary game wager, determined independent of the player's primary game wager, determined based on time (such as the time of day), determined independent of time (such as the time of day), determined based on an amount of coin-in accumulated in one or more pools, determined independent of an amount of coin-in accumulated in one or more pools, determined based on a status of the player (i.e., a player tracking status), determined independent of a status of the player (i.e., a player tracking status), determined based on one or more other determinations disclosed herein, deter-

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mined independent of any other determination disclosed herein or determined based on any other suitable method or criteria.

It should be appreciated from the above that various embodiments of the present disclosure provide a gaming system configured to: (1) display a first sequence triggering event, (2) display a first sequence comprising plays of a game, (3) display a plurality of accumulated symbol displays that are generated during the first sequence, each accumulated symbol display comprising a quantity of different symbols, wherein two of the accumulated symbol displays comprises different quantities of symbols, (4) display the quantity of different symbols of one of the accumulated symbol displays changing during the first sequence, (5) display a triggering event for a second sequence, and (6) display the second sequence wherein one of the accumulated symbol displays remaining after the triggering event for the second sequence occurs is activated and displays one of the symbols on said accumulated symbol display. In various such embodiments, the gaming system is configured to display the quantity of accumulated symbol displays changing during the first sequence. In various such embodiments, the gaming system is configured to display an additional accumulated symbol display added to the displayed accumulated symbol displays during the first sequence. In various such embodiments, the gaming system is configured to display one of the accumulated symbol displays removed from the displayed accumulated symbol displays during the first sequence. In various such embodiments, the gaming system is configured to display two of the accumulated symbol displays changing into a combined accumulated symbol display during the first sequence. In various such embodiments, the gaming system is configured to display the combined accumulated symbol display comprising a sum of the quantities of symbols of the two accumulated symbol displays that change into the combined accumulated symbol display. In various such embodiments, the gaming system is configured to display one of the accumulated symbol displays being split into two accumulated symbol displays during the first sequence. In various such embodiments, the gaming system is configured to display the two accumulated symbol displays that are split from the one accumulated symbol display comprise a total quantity of symbols equal to the quantity of symbols of the one accumulated symbol display. In various such embodiments, the gaming system is configured to display the second sequence wherein all of the accumulated symbol displays are activated and each accumulated symbol display displays one of the symbols of said accumulated symbol display. In various such embodiments, the gaming system is configured to display device an initial accumulated symbol display in association with the display of the first sequence triggering event. In various such embodiments, the gaming system is configured to display each of an initial quantity of accumulated symbol displays in association with the display of the first sequence triggering event, and for each initial accumulated symbol display a quantity of symbols on said accumulated symbol display. In various such embodiments, the gaming system is configured to display, for each of a plurality of the plays of the game of the first sequence, a randomly determined plurality of symbols, wherein one of the plurality of symbols causes a change to the accumulated symbol displays for the first sequence.

It should further be appreciated from the above that various embodiments of the present disclosure provide a gaming system configured to: (1) display a first sequence triggering event, (2) display each of an initial quantity of

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accumulated symbol displays after the display of the first sequence triggering event, and for each said accumulated symbol display a quantity of symbols on said accumulated symbol display, the quantity being at least one, (3) display a first sequence comprising plays of a game, (4) display the accumulated symbol displays changing during the first sequence, and which comprises a display of the quantity of different symbols on one of the accumulated symbol displays changing during the first sequence, and which comprises a display of the quantity of the accumulated symbol displays changing during the first sequence, (5) display a triggering event for a second sequence, and (6) display the second sequence wherein each of the accumulated symbol displays remaining after the triggering event for the second sequence occurs is activated and displays one of the symbols on said accumulated symbol display. In various such embodiments, the gaming system is configured to display an additional accumulated symbol display to be added to the displayed accumulated symbol displays during the first sequence. In various such embodiments, the gaming system is configured to display one of the accumulated symbol displays to be removed from the displayed accumulated symbol displays during the first sequence.

It should further be appreciated from the above that various embodiments of the present disclosure provide a gaming system configured to: (1) display a first sequence triggering event, (2) display a first sequence comprising plays of a game, (3) display a plurality of accumulated symbol displays that are displayed during the first sequence, each of the accumulated symbol displays comprising a quantity of different symbols, and two of the accumulated symbol displays comprising different quantities of symbols, (4) display two of the accumulated symbol displays being formed into a combined accumulated symbol display during the first sequence, wherein the combined accumulated symbol display comprises a sum of the quantities of symbols of the two accumulated symbol displays that are formed into the combined accumulated symbol display, (5) display one of the accumulated symbol displays being split into two accumulated symbol displays during the first sequence, wherein the two accumulated symbol displays that are split from the one accumulated symbol display comprise a total quantity of symbols equal to the quantity of symbols of said one accumulated symbol display, (6) display a triggering event for a second sequence, and (7) display the second sequence wherein each of the accumulated symbol displays remaining after the triggering event for the second sequence occurs is activated and displays one of the symbols on said accumulated symbol display. In various such embodiments, the gaming system is configured to display an initial accumulated symbol display in association with the display of the first sequence triggering event. In various such embodiments, the gaming system is configured to display each of an initial quantity of accumulated symbol displays in association with the display of the first sequence triggering event, and for each of said accumulated symbol display a quantity of symbols on said accumulated symbol display. In various such embodiments, the gaming system is configured to display, for each of a plurality of plays of the game of the first sequence, a randomly determined plurality of symbols, wherein one of the plurality of symbols causes a change to the accumulated symbol displays during the first sequence. In various such embodiments, the gaming system is configured to display an additional accumulated symbol display to be added to the accumulated symbol displays during the first sequence.

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It should further be appreciated from the above that the present disclosure provides technological improvements in gaming environments that include but are not limited to: (1) improvements in the displays of gaming systems; (2) improvements to the interaction between different sequences displayed by gaming systems; (3) improvements to symbol display accumulation sequences; (4) improvements to accumulated symbol display use sequences; (5) improvements to the integration and use of different symbol displays for gaming systems; (6) improvements to how symbol displays are changed by gaming systems; (7) improvements to how player interact with and change symbol displays of gaming systems; and (8) improvements to the use of dice in gaming systems.

Gaming Systems

The above-described embodiments of the present disclosure may be implemented in accordance with or in conjunction with one or more of a variety of different types of gaming systems, such as, but not limited to, those described below.

The present disclosure contemplates a variety of different gaming systems each having one or more of a plurality of different features, attributes, or characteristics. A “gaming system” as used herein refers to various configurations of: (a) one or more central servers, central controllers, or remote hosts; (b) one or more electronic gaming machines such as those located on a casino floor; and/or (c) one or more personal gaming devices, such as desktop computers, laptop computers, tablet computers or computing devices, personal digital assistants, mobile phones, and other mobile computing devices.

Thus, in various embodiments, the gaming system of the present disclosure includes: (a) one or more electronic gaming machines in combination with one or more central servers, central controllers, or remote hosts; (b) one or more personal gaming devices in combination with one or more central servers, central controllers, or remote hosts; (c) one or more personal gaming devices in combination with one or more electronic gaming machines; (d) one or more personal gaming devices, one or more electronic gaming machines, and one or more central servers, central controllers, or remote hosts in combination with one another; (e) a single electronic gaming machine; (f) a plurality of electronic gaming machines in combination with one another; (g) a single personal gaming device; (h) a plurality of personal gaming devices in combination with one another; (i) a single central server, central controller, or remote host; and/or (j) a plurality of central servers, central controllers, or remote hosts in combination with one another.

For brevity and clarity and unless specifically stated otherwise, the term “EGM” is used herein to refer to an electronic gaming machine (such as a slot machine, a video poker machine, a video lottery terminal (VLT), a video keno machine, or a video bingo machine located on a casino floor). Additionally, for brevity and clarity and unless specifically stated otherwise, “EGM” as used herein represents one EGM or a plurality of EGMs, “personal gaming device” as used herein represents one personal gaming device or a plurality of personal gaming devices, and “central server, central controller, or remote host” as used herein represents one central server, central controller, or remote host or a plurality of central servers, central controllers, or remote hosts.

As noted above, in various embodiments, the gaming system includes an EGM (or personal gaming device) in

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combination with a central server, central controller, or remote host. In such embodiments, the EGM (or personal gaming device) is configured to communicate with the central server, central controller, or remote host through a data network or remote communication link. In certain such embodiments, the EGM (or personal gaming device) is configured to communicate with another EGM (or personal gaming device) through the same data network or remote communication link or through a different data network or remote communication link. For example, the gaming system includes a plurality of EGMs that are each configured to communicate with a central server, central controller, or remote host through a data network.

In certain embodiments in which the gaming system includes an EGM (or personal gaming device) in combination with a central server, central controller, or remote host, the central server, central controller, or remote host is any suitable computing device (such as a server) that includes at least one processor and at least one memory device or data storage device. As further described herein, the EGM (or personal gaming device) includes at least one EGM (or personal gaming device) processor configured to transmit and receive data or signals representing events, messages, commands, or any other suitable information between the EGM (or personal gaming device) and the central server, central controller, or remote host. The at least one processor of that EGM (or personal gaming device) is configured to execute the events, messages, or commands represented by such data or signals in conjunction with the operation of the EGM (or personal gaming device). Moreover, the at least one processor of the central server, central controller, or remote host is configured to transmit and receive data or signals representing events, messages, commands, or any other suitable information between the central server, central controller, or remote host and the EGM (or personal gaming device). The at least one processor of the central server, central controller, or remote host is configured to execute the events, messages, or commands represented by such data or signals in conjunction with the operation of the central server, central controller, or remote host. One, more than one, or each of the functions of the central server, central controller, or remote host may be performed by the at least one processor of the EGM (or personal gaming device). Further, one, more than one, or each of the functions of the at least one processor of the EGM (or personal gaming device) may be performed by the at least one processor of the central server, central controller, or remote host.

In certain such embodiments, computerized instructions for controlling any games (such as any primary or base games and/or any secondary or bonus games) displayed by the EGM (or personal gaming device) are executed by the central server, central controller, or remote host. In such “thin client” embodiments, the central server, central controller, or remote host remotely controls any games (or other suitable interfaces) displayed by the EGM (or personal gaming device), and the EGM (or personal gaming device) is utilized to display such games (or suitable interfaces) and to receive one or more inputs or commands. In other such embodiments, computerized instructions for controlling any games displayed by the EGM (or personal gaming device) are communicated from the central server, central controller, or remote host to the EGM (or personal gaming device) and are stored in at least one memory device of the EGM (or personal gaming device). In such “thick client” embodiments, the at least one processor of the EGM (or personal gaming device) executes the computerized instructions to

control any games (or other suitable interfaces) displayed by the EGM (or personal gaming device).

In various embodiments in which the gaming system includes a plurality of EGMs (or personal gaming devices), one or more of the EGMs (or personal gaming devices) are thin client EGMs (or personal gaming devices) and one or more of the EGMs (or personal gaming devices) are thick client EGMs (or personal gaming devices). In other embodiments in which the gaming system includes one or more EGMs (or personal gaming devices), certain functions of one or more of the EGMs (or personal gaming devices) are implemented in a thin client environment, and certain other functions of one or more of the EGMs (or personal gaming devices) are implemented in a thick client environment. In one such embodiment in which the gaming system includes an EGM (or personal gaming device) and a central server, central controller, or remote host, computerized instructions for controlling any primary or base games displayed by the EGM (or personal gaming device) are communicated from the central server, central controller, or remote host to the EGM (or personal gaming device) in a thick client configuration, and computerized instructions for controlling any secondary or bonus games or other functions displayed by the EGM (or personal gaming device) are executed by the central server, central controller, or remote host in a thin client configuration.

In certain embodiments in which the gaming system includes: (a) an EGM (or personal gaming device) configured to communicate with a central server, central controller, or remote host through a data network; and/or (b) a plurality of EGMs (or personal gaming devices) configured to communicate with one another through a data network, the data network is a local area network (LAN) in which the EGMs (or personal gaming devices) are located substantially proximate to one another and/or the central server, central controller, or remote host. In one example, the EGMs (or personal gaming devices) and the central server, central controller, or remote host are located in a gaming establishment or a portion of a gaming establishment.

In other embodiments in which the gaming system includes: (a) an EGM (or personal gaming device) configured to communicate with a central server, central controller, or remote host through a data network; and/or (b) a plurality of EGMs (or personal gaming devices) configured to communicate with one another through a data network, the data network is a wide area network (WAN) in which one or more of the EGMs (or personal gaming devices) are not necessarily located substantially proximate to another one of the EGMs (or personal gaming devices) and/or the central server, central controller, or remote host. For example, one or more of the EGMs (or personal gaming devices) are located: (a) in an area of a gaming establishment different from an area of the gaming establishment in which the central server, central controller, or remote host is located; or (b) in a gaming establishment different from the gaming establishment in which the central server, central controller, or remote host is located. In another example, the central server, central controller, or remote host is not located within a gaming establishment in which the EGMs (or personal gaming devices) are located. In certain embodiments in which the data network is a WAN, the gaming system includes a central server, central controller, or remote host and an EGM (or personal gaming device) each located in a different gaming establishment in a same geographic area, such as a same city or a same state. Gaming systems in which the data network is a WAN are substantially identical to gaming systems in which the data network is a LAN,

though the quantity of EGMs (or personal gaming devices) in such gaming systems may vary relative to one another.

In further embodiments in which the gaming system includes: (a) an EGM (or personal gaming device) configured to communicate with a central server, central controller, or remote host through a data network; and/or (b) a plurality of EGMs (or personal gaming devices) configured to communicate with one another through a data network, the data network is an internet (such as the Internet) or an intranet. In certain such embodiments, an Internet browser of the EGM (or personal gaming device) is usable to access an Internet game page from any location where an Internet connection is available. In one such embodiment, after the EGM (or personal gaming device) accesses the Internet game page, the central server, central controller, or remote host identifies a player before enabling that player to place any wagers on any plays of any wagering games. In one example, the central server, central controller, or remote host identifies the player by requiring a player account of the player to be logged into via an input of a unique username and password combination assigned to the player. The central server, central controller, or remote host may, however, identify the player in any other suitable manner, such as by validating a player tracking identification number associated with the player; by reading a player tracking card or other smart card inserted into a card reader (as described below); by validating a unique player identification number associated with the player by the central server, central controller, or remote host; or by identifying the EGM (or personal gaming device), such as by identifying the MAC address or the IP address of the Internet facilitator. In various embodiments, once the central server, central controller, or remote host identifies the player, the central server, central controller, or remote host enables placement of one or more wagers on one or more plays of one or more primary or base games and/or one or more secondary or bonus games, and displays those plays via the Internet browser of the EGM (or personal gaming device). Examples of implementations of Internet-based gaming are further described in U.S. Pat. No. 8,764,566, entitled "Internet Remote Game Server," and U.S. Pat. No. 8,147,334, entitled "Universal Game Server," which are incorporated herein by reference.

The central server, central controller, or remote host and the EGM (or personal gaming device) are configured to connect to the data network or remote communications link in any suitable manner. In various embodiments, such a connection is accomplished via a conventional phone line or other data transmission line, a digital subscriber line (DSL), a T-1 line, a coaxial cable, a fiber optic cable, a wireless or wired routing device, a mobile communications network connection (such as a cellular network or mobile Internet network), or any other suitable medium. The expansion in the quantity of computing devices and the quantity and speed of Internet connections in recent years increases opportunities for players to use a variety of EGMs (or personal gaming devices) to play games from an ever-increasing quantity of remote sites. Additionally, the enhanced bandwidth of digital wireless communications may render such technology suitable for some or all communications, particularly if such communications are encrypted. Higher data transmission speeds may be useful for enhancing the sophistication and response of the display and interaction with players.

EGM Components

FIG. 4 is a block diagram of an example EGM 1000 and FIGS. 5A and 5B include two different example EGMs

2000a and **2000b**. The EGMs **1000**, **2000a**, and **2000b** are merely example EGMs, and different EGMs may be implemented using different combinations of the components shown in the EGMs **1000**, **2000a**, and **2000b**. Although the below refers to EGMs, in various embodiments personal gaming devices (such as personal gaming device **2000c** of FIG. 5C) may include some or all of the below components.

In these embodiments, the EGM **1000** includes a master gaming controller **1012** configured to communicate with and to operate with a plurality of peripheral devices **1022**.

The master gaming controller **1012** includes at least one processor **1010**. The at least one processor **1010** is any suitable processing device or set of processing devices, such as a microprocessor, a microcontroller-based platform, a suitable integrated circuit, or one or more application-specific integrated circuits (ASICs), configured to execute software enabling various configuration and reconfiguration tasks, such as: (1) communicating with a remote source (such as a server that stores authentication information or game information) via a communication interface **1006** of the master gaming controller **1012**; (2) converting signals read by an interface to a format corresponding to that used by software or memory of the EGM; (3) accessing memory to configure or reconfigure game parameters in the memory according to indicia read from the EGM; (4) communicating with interfaces and the peripheral devices **1022** (such as input/output devices); and/or (5) controlling the peripheral devices **1022**. In certain embodiments, one or more components of the master gaming controller **1012** (such as the at least one processor **1010**) reside within a housing of the EGM (described below), while in other embodiments at least one component of the master gaming controller **1012** resides outside of the housing of the EGM.

The master gaming controller **1012** also includes at least one memory device **1016**, which includes: (1) volatile memory (e.g., RAM **1009**, which can include non-volatile RAM, magnetic RAM, ferroelectric RAM, and any other suitable forms); (2) non-volatile memory **1019** (e.g., disk memory, FLASH memory, EPROMs, EEPROMs, memristor-based non-volatile solid-state memory, etc.); (3) unalterable memory (e.g., EPROMs **1008**); (4) read-only memory; and/or (5) a secondary memory storage device **1015**, such as a non-volatile memory device, configured to store gaming software related information (the gaming software related information and the memory may be used to store various audio files and games not currently being used and invoked in a configuration or reconfiguration). Any other suitable magnetic, optical, and/or semiconductor memory may operate in conjunction with the EGM disclosed herein. In certain embodiments, the at least one memory device **1016** resides within the housing of the EGM (described below), while in other embodiments at least one component of the at least one memory device **1016** resides outside of the housing of the EGM.

The at least one memory device **1016** is configured to store, for example: (1) configuration software **1014**, such as all the parameters and settings for a game playable on the EGM; (2) associations **1018** between configuration indicia read from an EGM with one or more parameters and settings; (3) communication protocols configured to enable the at least one processor **1010** to communicate with the peripheral devices **1022**; and/or (4) communication transport protocols (such as TCP/IP, USB, Firewire, IEEE1394, Bluetooth, IEEE 802.11x (IEEE 802.11 standards), hipervlan/2, HomeRF, etc.) configured to enable the EGM to communicate with local and non-local devices using such protocols. In one implementation, the master gaming controller **1012**

communicates with other devices using a serial communication protocol. A few non-limiting examples of serial communication protocols that other devices, such as peripherals (e.g., a bill validator or a ticket printer), may use to communicate with the master game controller **1012** include USB, RS-232, and Netplex (a proprietary protocol developed by IGT).

In certain embodiments, the at least one memory device **1016** is configured to store program code and instructions executable by the at least one processor of the EGM to control the EGM. The at least one memory device **1016** of the EGM also stores other operating data, such as image data, event data, input data, random number generators (RNGs) or pseudo-RNGs, payable data or information, and/or applicable game rules that relate to the play of one or more games on the EGM. In various embodiments, part or all of the program code and/or the operating data described above is stored in at least one detachable or removable memory device including, but not limited to, a cartridge, a disk, a CD ROM, a DVD, a USB memory device, or any other suitable non-transitory computer readable medium. In certain such embodiments, an operator (such as a gaming establishment operator) and/or a player uses such a removable memory device in an EGM to implement at least part of the present disclosure. In other embodiments, part or all of the program code and/or the operating data is downloaded to the at least one memory device of the EGM through any suitable data network described above (such as an Internet or intranet).

The at least one memory device **1016** also stores a plurality of device drivers **1042**. Examples of different types of device drivers include device drivers for EGM components and device drivers for the peripheral components **1022**. Typically, the device drivers **1042** utilize various communication protocols that enable communication with a particular physical device. The device driver abstracts the hardware implementation of that device. For example, a device driver may be written for each type of card reader that could potentially be connected to the EGM. Non-limiting examples of communication protocols used to implement the device drivers include Netplex, USB, Serial, Ethernet **175**, Firewire, I/O debouncer, direct memory map, serial, PCI, parallel, RF, Bluetooth™, near-field communications (e.g., using near-field magnetics), 802.11 (WiFi), etc. In one embodiment, when one type of a particular device is exchanged for another type of the particular device, the at least one processor of the EGM loads the new device driver from the at least one memory device to enable communication with the new device. For instance, one type of card reader in the EGM can be replaced with a second different type of card reader when device drivers for both card readers are stored in the at least one memory device.

In certain embodiments, the software units stored in the at least one memory device **1016** can be upgraded as needed. For instance, when the at least one memory device **1016** is a hard drive, new games, new game options, new parameters, new settings for existing parameters, new settings for new parameters, new device drivers, and new communication protocols can be uploaded to the at least one memory device **1016** from the master game controller **1012** or from some other external device. As another example, when the at least one memory device **1016** includes a CD/DVD drive including a CD/DVD configured to store game options, parameters, and settings, the software stored in the at least one memory device **1016** can be upgraded by replacing a first CD/DVD with a second CD/DVD. In yet another example, when the at least one memory device **1016** uses

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flash memory **1019** or EPROM **1008** units configured to store games, game options, parameters, and settings, the software stored in the flash and/or EPROM memory units can be upgraded by replacing one or more memory units with new memory units that include the upgraded software. In another embodiment, one or more of the memory devices, such as the hard drive, may be employed in a game software download process from a remote software server.

In some embodiments, the at least one memory device **1016** also stores authentication and/or validation components **1044** configured to authenticate/validate specified EGM components and/or information, such as hardware components, software components, firmware components, peripheral device components, user input device components, information received from one or more user input devices, information stored in the at least one memory device **1016**, etc. Examples of various authentication and/or validation components are described in U.S. Pat. No. 6,620,047, entitled "Electronic Gaming Apparatus Having Authentication Data Sets."

In certain embodiments, the peripheral devices **1022** include several device interfaces, such as: (1) at least one output device **1020** including at least one display device **1035**; (2) at least one input device **1030** (which may include contact and/or non-contact interfaces); (3) at least one transponder **1054**; (4) at least one wireless communication component **1056**; (5) at least one wired/wireless power distribution component **1058**; (6) at least one sensor **1060**; (7) at least one data preservation component **1062**; (8) at least one motion/gesture analysis and interpretation component **1064**; (9) at least one motion detection component **1066**; (10) at least one portable power source **1068**; (11) at least one geolocation module **1076**; (12) at least one user identification module **1077**; (13) at least one player/device tracking module **1078**; and (14) at least one information filtering module **1079**.

The at least one output device **1020** includes at least one display device **1035** configured to display any game(s) displayed by the EGM and any suitable information associated with such game(s). In certain embodiments, the display devices are connected to or mounted on a housing of the EGM (described below). In various embodiments, the display devices serve as digital glass configured to advertise certain games or other aspects of the gaming establishment in which the EGM is located. In various embodiments, the EGM includes one or more of the following display devices: (a) a central display device; (b) a player tracking display configured to display various information regarding a player's player tracking status (as described below); (c) a secondary or upper display device in addition to the central display device and the player tracking display; (d) a credit display configured to display a current quantity of credits, amount of cash, account balance, or the equivalent; and (e) a bet display configured to display an amount wagered for one or more plays of one or more games. The example EGM **2000a** illustrated in FIG. 5A includes a central display device **2116**, a player tracking display **2140**, a credit display **2120**, and a bet display **2122**. The example EGM **2000b** illustrated in FIG. 5B includes a central display device **2116**, an upper display device **2118**, a player tracking display **2140**, a credit display **2120**, and a bet display **2122**.

In various embodiments, the display devices include, without limitation: a monitor, a television display, a plasma display, a liquid crystal display (LCD), a display based on light emitting diodes (LEDs), a display based on a plurality of organic light-emitting diodes (OLEDs), a display based on polymer light-emitting diodes (PLEDs), a display based

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on a plurality of surface-conduction electron-emitters (SEDs), a display including a projected and/or reflected image, or any other suitable electronic device or display mechanism. In certain embodiments, as described above, the display device includes a touch-screen with an associated touch-screen controller. The display devices may be of any suitable sizes, shapes, and configurations.

The display devices of the EGM are configured to display one or more game and/or non-game images, symbols, and indicia. In certain embodiments, the display devices of the EGM are configured to display any suitable visual representation or exhibition of the movement of objects; dynamic lighting; video images; images of people, characters, places, things, and faces of cards; and the like. In certain embodiments, the display devices of the EGM are configured to display one or more video reels, one or more video wheels, and/or one or more video dice. In other embodiments, certain of the displayed images, symbols, and indicia are in mechanical form. That is, in these embodiments, the display device includes any electromechanical device, such as one or more rotatable wheels, one or more reels, and/or one or more dice, configured to display at least one or a plurality of game or other suitable images, symbols, or indicia.

In various embodiments, the at least one output device **1020** includes a payout device. In these embodiments, after the EGM receives an actuation of a cashout device (described below), the EGM causes the payout device to provide a payment to the player. In one embodiment, the payout device is one or more of: (a) a ticket printer and dispenser configured to print and dispense a ticket or credit slip associated with a monetary value, wherein the ticket or credit slip may be redeemed for its monetary value via a cashier, a kiosk, or other suitable redemption system; (b) a bill dispenser configured to dispense paper currency; (c) a coin dispenser configured to dispense coins or tokens (such as into a coin payout tray); and (d) any suitable combination thereof. The example EGMs **2000a** and **2000b** illustrated in FIGS. 5A and 5B each include a ticket printer and dispenser **2136**. Examples of ticket-in ticket-out (TITO) technology are described in U.S. Pat. No. 5,429,361, entitled "Gaming Machine Information, Communication and Display System"; U.S. Pat. No. 5,470,079, entitled "Gaming Machine Accounting and Monitoring System"; U.S. Pat. No. 5,265,874, entitled "Cashless Gaming Apparatus and Method"; U.S. Pat. No. 6,729,957, entitled "Gaming Method and Host Computer with Ticket-In/Ticket-Out Capability"; U.S. Pat. No. 6,729,958, entitled "Gaming System with Ticket-In/Ticket-Out Capability"; U.S. Pat. No. 6,736,725, entitled "Gaming Method and Host Computer with Ticket-In/Ticket-Out Capability"; U.S. Pat. No. 7,275,991, entitled "Slot Machine with Ticket-In/Ticket-Out Capability"; U.S. Pat. No. 6,048,269, entitled "Coinless Slot Machine System and Method"; and U.S. Pat. No. 5,290,003, entitled "Gaming Machine and Coupons."

In certain embodiments, rather than dispensing bills, coins, or a physical ticket having a monetary value to the player following receipt of an actuation of the cashout device, the payout device is configured to cause a payment to be provided to the player in the form of an electronic funds transfer, such as via a direct deposit into a bank account, a casino account, or a prepaid account of the player; via a transfer of funds onto an electronically recordable identification card or smart card of the player; or via sending a virtual ticket having a monetary value to an electronic device of the player. Examples of providing payment using

virtual tickets are described in U.S. Pat. No. 8,613,659, entitled "Virtual Ticket-In and Ticket-Out on a Gaming Machine."

While any credit balances, any wagers, any values, and any awards are described herein as amounts of monetary credits or currency, one or more of such credit balances, such wagers, such values, and such awards may be for non-monetary credits, promotional credits, of player tracking points or credits.

In certain embodiments, the at least one output device **1020** is a sound generating device controlled by one or more sound cards. In one such embodiment, the sound generating device includes one or more speakers or other sound generating hardware and/or software configured to generate sounds, such as by playing music for any games or by playing music for other modes of the EGM, such as an attract mode. The example EGMs **2000a** and **2000b** illustrated in FIGS. **5A** and **5B** each include a plurality of speakers **2150**. In another such embodiment, the EGM provides dynamic sounds coupled with attractive multimedia images displayed on one or more of the display devices to provide an audio-visual representation or to otherwise display full-motion video with sound to attract players to the EGM. In certain embodiments, the EGM displays a sequence of audio and/or visual attraction messages during idle periods to attract potential players to the EGM. The videos may be customized to provide any appropriate information.

The at least one input device **1030** may include any suitable device that enables an input signal to be produced and received by the at least one processor **1010** of the EGM.

In one embodiment, the at least one input device **1030** includes a payment device configured to communicate with the at least one processor of the EGM to fund the EGM. In certain embodiments, the payment device includes one or more of: (a) a bill acceptor into which paper money is inserted to fund the EGM; (b) a ticket acceptor into which a ticket or a voucher is inserted to fund the EGM; (c) a coin slot into which coins or tokens are inserted to fund the EGM; (d) a reader or a validator for credit cards, debit cards, or credit slips into which a credit card, debit card, or credit slip is inserted to fund the EGM; (e) a player identification card reader into which a player identification card is inserted to fund the EGM; or (f) any suitable combination thereof. The example EGMs **2000a** and **2000b** illustrated in FIGS. **5A** and **5B** each include a combined bill and ticket acceptor **2128** and a coin slot **2126**.

In one embodiment, the at least one input device **1030** includes a payment device configured to enable the EGM to be funded via an electronic funds transfer, such as a transfer of funds from a bank account. In another embodiment, the EGM includes a payment device configured to communicate with a mobile device of a player, such as a mobile phone, a radio frequency identification tag, or any other suitable wired or wireless device, to retrieve relevant information associated with that player to fund the EGM. Examples of funding an EGM via communication between the EGM and a mobile device (such as a mobile phone) of a player are described in U.S. Patent Application Publication No. 2013/0344942, entitled "Avatar as Security Measure for Mobile Device Use with Electronic Gaming Machine," which is incorporated herein by reference. When the EGM is funded, the at least one processor determines the amount of funds entered and displays the corresponding amount on a credit display or any other suitable display as described below.

In certain embodiments, the at least one input device **1030** includes at least one wagering or betting device. In various

embodiments, the one or more wagering or betting devices are each: (1) a mechanical button supported by the housing of the EGM (such as a hard key or a programmable soft key), or (2) an icon displayed on a display device of the EGM (described below) that is actuatable via a touch screen of the EGM (described below) or via use of a suitable input device of the EGM (such as a mouse or a joystick). One such wagering or betting device is as a maximum wager or bet device that, when actuated, causes the EGM to place a maximum wager on a play of a game. Another such wagering or betting device is a repeat bet device that, when actuated, causes the EGM to place a wager that is equal to the previously-placed wager on a play of a game. A further such wagering or betting device is a bet one device that, when actuated, causes the EGM to increase the wager by one credit. Generally, upon actuation of one of the wagering or betting devices, the quantity of credits displayed in a credit meter (described below) decreases by the amount of credits wagered, while the quantity of credits displayed in a bet display (described below) increases by the amount of credits wagered.

In various embodiments, the at least one input device **1030** includes at least one game play activation device. In various embodiments, the one or more game play initiation devices are each: (1) a mechanical button supported by the housing of the EGM (such as a hard key or a programmable soft key), or (2) an icon displayed on a display device of the EGM (described below) that is actuatable via a touch screen of the EGM (described below) or via use of a suitable input device of the EGM (such as a mouse or a joystick). After a player appropriately funds the EGM and places a wager, the EGM activates the game play activation device to enable the player to actuate the game play activation device to initiate a play of a game on the EGM (or another suitable sequence of events associated with the EGM). After the EGM receives an actuation of the game play activation device, the EGM initiates the play of the game. The example EGMs **2000a** and **2000b** illustrated in FIGS. **5A** and **5B** each include a game play activation device in the form of a game play initiation button **2132**. In other embodiments, the EGM begins game play automatically upon appropriate funding rather than upon utilization of the game play activation device.

In other embodiments, the at least one input device **1030** includes a cashout device. In various embodiments, the cashout device is: (1) a mechanical button supported by the housing of the EGM (such as a hard key or a programmable soft key), or (2) an icon displayed on a display device of the EGM (described below) that is actuatable via a touch screen of the EGM (described below) or via use of a suitable input device of the EGM (such as a mouse or a joystick). When the EGM receives an actuation of the cashout device from a player and the player has a positive (i.e., greater-than-zero) credit balance, the EGM initiates a payout associated with the player's credit balance. The example EGMs **2000a** and **2000b** illustrated in FIGS. **5A** and **5B** each include a cashout device in the form of a cashout button **2134**.

In various embodiments, the at least one input device **1030** includes a plurality of buttons that are programmable by the EGM operator to, when actuated, cause the EGM to perform particular functions. For instance, such buttons may be hard keys, programmable soft keys, or icons icon displayed on a display device of the EGM (described below) that are actuatable via a touch screen of the EGM (described below) or via use of a suitable input device of the EGM (such as a mouse or a joystick). The example EGMs **2000a**

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and **2000b** illustrated in FIGS. **5A** and **5B** each include a plurality of such buttons **2130**.

In certain embodiments, the at least one input device **1030** includes a touch-screen coupled to a touch-screen controller or other touch-sensitive display overlay to enable interaction with any images displayed on a display device (as described below). One such input device is a conventional touch-screen button panel. The touch-screen and the touch-screen controller are connected to a video controller. In these embodiments, signals are input to the EGM by touching the touch screen at the appropriate locations.

In embodiments including a player tracking system, as further described below, the at least one input device **1030** includes a card reader in communication with the at least one processor of the EGM. The example EGMs **2000a** and **2000b** illustrated in FIGS. **5A** and **5B** each include a card reader **2138**. The card reader is configured to read a player identification card inserted into the card reader.

The at least one wireless communication component **1056** includes one or more communication interfaces having different architectures and utilizing a variety of protocols, such as (but not limited to) 802.11 (WiFi); 802.15 (including Bluetooth™); 802.16 (WiMax); 802.22; cellular standards such as CDMA, CDMA2000, and WCDMA; Radio Frequency (e.g., RFID); infrared; and Near Field Magnetic communication protocols. The at least one wireless communication component **1056** transmits electrical, electromagnetic, or optical signals that carry digital data streams or analog signals representing various types of information.

The at least one wired/wireless power distribution component **1058** includes components or devices that are configured to provide power to other devices. For example, in one embodiment, the at least one power distribution component **1058** includes a magnetic induction system that is configured to provide wireless power to one or more user input devices near the EGM. In one embodiment, a user input device docking region is provided, and includes a power distribution component that is configured to recharge a user input device without requiring metal-to-metal contact. In one embodiment, the at least one power distribution component **1058** is configured to distribute power to one or more internal components of the EGM, such as one or more rechargeable power sources (e.g., rechargeable batteries) located at the EGM.

In certain embodiments, the at least one sensor **1060** includes at least one of: optical sensors, pressure sensors, RF sensors, infrared sensors, image sensors, thermal sensors, and biometric sensors. The at least one sensor **1060** may be used for a variety of functions, such as: detecting movements and/or gestures of various objects within a predetermined proximity to the EGM; detecting the presence and/or identity of various persons (e.g., players, casino employees, etc.), devices (e.g., user input devices), and/or systems within a predetermined proximity to the EGM.

The at least one data preservation component **1062** is configured to detect or sense one or more events and/or conditions that, for example, may result in damage to the EGM and/or that may result in loss of information associated with the EGM. Additionally, the data preservation system **1062** may be operable to initiate one or more appropriate action(s) in response to the detection of such events/conditions.

The at least one motion/gesture analysis and interpretation component **1064** is configured to analyze and/or interpret information relating to detected player movements and/or gestures to determine appropriate player input information relating to the detected player movements and/or gestures.

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For example, in one embodiment, the at least one motion/gesture analysis and interpretation component **1064** is configured to perform one or more of the following functions: analyze the detected gross motion or gestures of a player; interpret the player's motion or gestures (e.g., in the context of a casino game being played) to identify instructions or input from the player; utilize the interpreted instructions/input to advance the game state; etc. In other embodiments, at least a portion of these additional functions may be implemented at a remote system or device.

The at least one portable power source **1068** enables the EGM to operate in a mobile environment. For example, in one embodiment, the EGM **300** includes one or more rechargeable batteries.

The at least one geolocation module **1076** is configured to acquire geolocation information from one or more remote sources and use the acquired geolocation information to determine information relating to a relative and/or absolute position of the EGM. For example, in one implementation, the at least one geolocation module **1076** is configured to receive GPS signal information for use in determining the position or location of the EGM. In another implementation, the at least one geolocation module **1076** is configured to receive multiple wireless signals from multiple remote devices (e.g., EGMs, servers, wireless access points, etc.) and use the signal information to compute position/location information relating to the position or location of the EGM.

The at least one user identification module **1077** is configured to determine the identity of the current user or current owner of the EGM. For example, in one embodiment, the current user is required to perform a login process at the EGM in order to access one or more features. Alternatively, the EGM is configured to automatically determine the identity of the current user based on one or more external signals, such as an RFID tag or badge worn by the current user and that provides a wireless signal to the EGM that is used to determine the identity of the current user. In at least one embodiment, various security features are incorporated into the EGM to prevent unauthorized users from accessing confidential or sensitive information.

The at least one information filtering module **1079** is configured to perform filtering (e.g., based on specified criteria) of selected information to be displayed at one or more displays **1035** of the EGM.

In various embodiments, the EGM includes a plurality of communication ports configured to enable the at least one processor of the EGM to communicate with and to operate with external peripherals, such as: accelerometers, arcade sticks, bar code readers, bill validators, biometric input devices, bonus devices, button panels, card readers, coin dispensers, coin hoppers, display screens or other displays or video sources, expansion buses, information panels, keypads, lights, mass storage devices, microphones, motion sensors, motors, printers, reels, SCSI ports, solenoids, speakers, thumbsticks, ticket readers, touch screens, trackballs, touchpads, wheels, and wireless communication devices. U.S. Pat. No. 7,290,072 describes a variety of EGMs including one or more communication ports that enable the EGMs to communicate and operate with one or more external peripherals.

As generally described above, in certain embodiments, such as the example EGMs **2000a** and **2000b** illustrated in FIGS. **5A** and **5B**, the EGM has a support structure, housing, or cabinet that provides support for a plurality of the input devices and the output devices of the EGM. Further, the EGM is configured such that a player may operate it while standing or sitting. In various embodiments, the EGM is

positioned on a base or stand, or is configured as a pub-style tabletop game (not shown) that a player may operate typically while sitting. As illustrated by the different example EGMs 2000a and 2000b shown in FIGS. 5A and 5B, EGMs may have varying housing and display configurations.

In certain embodiments, the EGM is a device that has obtained approval from a regulatory gaming commission, and in other embodiments, the EGM is a device that has not obtained approval from a regulatory gaming commission.

The EGMs described above are merely three examples of different types of EGMs. Certain of these example EGMs may include one or more elements that may not be included in all gaming systems, and these example EGMs may not include one or more elements that are included in other gaming systems. For example, certain EGMs include a coin acceptor while others do not.

Operation of Primary or Base Games and/or Secondary or Bonus Games

In various embodiments, an EGM may be implemented in one of a variety of different configurations. In various embodiments, the EGM may be implemented as one of: (a) a dedicated EGM in which computerized game programs executable by the EGM for controlling any primary or base games (referred to herein as “primary games”) and/or any secondary or bonus games or other functions (referred to herein as “secondary games”) displayed by the EGM are provided with the EGM before delivery to a gaming establishment or before being provided to a player; and (b) a changeable EGM in which computerized game programs executable by the EGM for controlling any primary games and/or secondary games displayed by the EGM are downloadable or otherwise transferred to the EGM through a data network or remote communication link; from a USB drive, flash memory card, or other suitable memory device; or in any other suitable manner after the EGM is physically located in a gaming establishment or after the EGM is provided to a player.

As generally explained above, in various embodiments in which the gaming system includes a central server, central controller, or remote host and a changeable EGM, the at least one memory device of the central server, central controller, or remote host stores different game programs and instructions executable by the at least one processor of the changeable EGM to control one or more primary games and/or secondary games displayed by the changeable EGM. More specifically, each such executable game program represents a different game or a different type of game that the at least one changeable EGM is configured to operate. In one example, certain of the game programs are executable by the changeable EGM to operate games having the same or substantially the same game play but different paytables. In different embodiments, each executable game program is associated with a primary game, a secondary game, or both. In certain embodiments, an executable game program is executable by the at least one processor of the at least one changeable EGM as a secondary game to be played simultaneously with a play of a primary game (which may be downloaded to or otherwise stored on the at least one changeable EGM), or vice versa.

In operation of such embodiments, the central server, central controller, or remote host is configured to communicate one or more of the stored executable game programs to the at least one processor of the changeable EGM. In different embodiments, a stored executable game program is communicated or delivered to the at least one processor of

the changeable EGM by: (a) embedding the executable game program in a device or a component (such as a microchip to be inserted into the changeable EGM); (b) writing the executable game program onto a disc or other media; or (c) uploading or streaming the executable game program over a data network (such as a dedicated data network). After the executable game program is communicated from the central server, central controller, or remote host to the changeable EGM, the at least one processor of the changeable EGM executes the executable game program to enable the primary game and/or the secondary game associated with that executable game program to be played using the display device(s) and/or the input device(s) of the changeable EGM. That is, when an executable game program is communicated to the at least one processor of the changeable EGM, the at least one processor of the changeable EGM changes the game or the type of game that may be played using the changeable EGM.

In certain embodiments, the gaming system randomly determines any game outcome(s) (such as a win outcome) and/or award(s) (such as a quantity of credits to award for the win outcome) for a play of a primary game and/or a play of a secondary game based on probability data. In certain such embodiments, this random determination is provided through utilization of an RNG, such as a true RNG or a pseudo RNG, or any other suitable randomization process. In one such embodiment, each game outcome or award is associated with a probability, and the gaming system generates the game outcome(s) and/or the award(s) to be provided based on the associated probabilities. In these embodiments, since the gaming system generates game outcomes and/or awards randomly or based on one or more probability calculations, there is no certainty that the gaming system will ever provide any specific game outcome and/or award.

In certain embodiments, the gaming system maintains one or more predetermined pools or sets of predetermined game outcomes and/or awards. In certain such embodiments, upon generation or receipt of a game outcome and/or award request, the gaming system independently selects one of the predetermined game outcomes and/or awards from the one or more pools or sets. The gaming system flags or marks the selected game outcome and/or award as used. Once a game outcome or an award is flagged as used, it is prevented from further selection from its respective pool or set; that is, the gaming system does not select that game outcome or award upon another game outcome and/or award request. The gaming system provides the selected game outcome and/or award. Examples of this type of award evaluation are described in U.S. Pat. No. 7,470,183, entitled “Finite Pool Gaming Method and Apparatus”; U.S. Pat. No. 7,563,163, entitled “Gaming Device Including Outcome Pools for Providing Game Outcomes”; U.S. Pat. No. 7,833,092, entitled “Method and System for Compensating for Player Choice in a Game of Chance”; U.S. Pat. No. 8,070,579, entitled “Bingo System with Downloadable Common Patterns”; and U.S. Pat. No. 8,398,472, entitled “Central Determination Poker Game.”

In certain embodiments, the gaming system determines a predetermined game outcome and/or award based on the results of a bingo, keno, or lottery game. In certain such embodiments, the gaming system utilizes one or more bingo, keno, or lottery games to determine the predetermined game outcome and/or award provided for a primary game and/or a secondary game. The gaming system is provided or associated with a bingo card. Each bingo card consists of a matrix or array of elements, wherein each element is designated with separate indicia. After a bingo card is provided,

the gaming system randomly selects or draws a plurality of the elements. As each element is selected, a determination is made as to whether the selected element is present on the bingo card. If the selected element is present on the bingo card, that selected element on the provided bingo card is marked or flagged. This process of selecting elements and marking any selected elements on the provided bingo cards continues until one or more predetermined patterns are marked on one or more of the provided bingo cards. After one or more predetermined patterns are marked on one or more of the provided bingo cards, game outcome and/or award is determined based, at least in part, on the selected elements on the provided bingo cards. Examples of this type of award determination are described in U.S. Pat. No. 7,753,774, entitled "Using Multiple Bingo Cards to Represent Multiple Slot Paylines and Other Class III Game Options"; U.S. Pat. No. 7,731,581, entitled "Multi-Player Bingo Game with Multiple Alternative Outcome Displays"; U.S. Pat. No. 7,955,170, entitled "Providing Non-Bingo Outcomes for a Bingo Game"; U.S. Pat. No. 8,070,579, entitled "Bingo System with Downloadable Common Patterns"; and U.S. Pat. No. 8,500,538, entitled "Bingo Gaming System and Method for Providing Multiple Outcomes from Single Bingo Pattern."

In certain embodiments in which the gaming system includes a central server, central controller, or remote host and an EGM, the EGM is configured to communicate with the central server, central controller, or remote host for monitoring purposes only. In such embodiments, the EGM determines the game outcome(s) and/or award(s) to be provided in any of the manners described above, and the central server, central controller, or remote host monitors the activities and events occurring on the EGM. In one such embodiment, the gaming system includes a real-time or online accounting and gaming information system configured to communicate with the central server, central controller, or remote host. In this embodiment, the accounting and gaming information system includes: (a) a player database configured to store player profiles, (b) a player tracking module configured to track players (as described below), and (c) a credit system configured to provide automated transactions. Examples of such accounting systems are described in U.S. Pat. No. 6,913,534, entitled "Gaming Machine Having a Lottery Game and Capability for Integration with Gaming Device Accounting System and Player Tracking System," and U.S. Pat. No. 8,597,116, entitled "Virtual Player Tracking and Related Services."

As noted above, in various embodiments, the gaming system includes one or more executable game programs executable by at least one processor of the gaming system to provide one or more primary games and one or more secondary games. The primary game(s) and the secondary game(s) may comprise any suitable games and/or wagering games, such as, but not limited to electro-mechanical or video slot or spinning reel type games; video card games such as video draw poker, multi-hand video draw poker, other video poker games, video blackjack games, and video baccarat games; video keno games; video bingo games; and video selection games.

In certain embodiments in which the primary game is a slot or spinning reel type game, the gaming system includes one or more reels in either an electromechanical form with mechanical rotating reels or in a video form with simulated reels and movement thereof. Each reel displays a plurality of indicia or symbols, such as bells, hearts, fruits, numbers, letters, bars, or other images that typically correspond to a theme associated with the gaming system. In certain such

embodiments, the gaming system includes one or more paylines associated with the reels. The example EGM 2000b shown in FIG. 5B includes a payline 1152 and a plurality of reels 1154. In certain embodiments, one or more of the reels are independent reels or unisymbol reels. In such embodiments, each independent reel generates and displays one symbol.

In various embodiments, one or more of the paylines is horizontal, vertical, circular, diagonal, angled, or any suitable combination thereof. In other embodiments, each of one or more of the paylines is associated with a plurality of adjacent symbol display areas on a requisite number of adjacent reels. In one such embodiment, one or more paylines are formed between at least two symbol display areas that are adjacent to each other by either sharing a common side or sharing a common corner (i.e., such paylines are connected paylines). The gaming system enables a wager to be placed on one or more of such paylines to activate such paylines. In other embodiments in which one or more paylines are formed between at least two adjacent symbol display areas, the gaming system enables a wager to be placed on a plurality of symbol display areas, which activates those symbol display areas.

In various embodiments, the gaming system provides one or more awards after a spin of the reels when specified types and/or configurations of the indicia or symbols on the reels occur on an active payline or otherwise occur in a winning pattern, occur on the requisite number of adjacent reels, and/or occur in a scatter pay arrangement.

In certain embodiments, the gaming system employs a ways to win award determination. In these embodiments, any outcome to be provided is determined based on a number of associated symbols that are generated in active symbol display areas on the requisite number of adjacent reels (i.e., not on paylines passing through any displayed winning symbol combinations). If a winning symbol combination is generated on the reels, one award for that occurrence of the generated winning symbol combination is provided. Examples of ways to win award determinations are described in U.S. Pat. No. 8,012,011, entitled "Gaming Device and Method Having Independent Reels and Multiple Ways of Winning"; U.S. Pat. No. 8,241,104, entitled "Gaming Device and Method Having Designated Rules for Determining Ways To Win"; and U.S. Pat. No. 8,430,739, entitled "Gaming System and Method Having Wager Dependent Different Symbol Evaluations."

In various embodiments, the gaming system includes a progressive award. Typically, a progressive award includes an initial amount and an additional amount funded through a portion of each wager placed to initiate a play of a primary game. When one or more triggering events occurs, the gaming system provides at least a portion of the progressive award. After the gaming system provides the progressive award, an amount of the progressive award is reset to the initial amount and a portion of each subsequent wager is allocated to the next progressive award. Examples of progressive gaming systems are described in U.S. Pat. No. 7,585,223, entitled "Server Based Gaming System Having Multiple Progressive Awards"; U.S. Pat. No. 7,651,392, entitled "Gaming Device System Having Partial Progressive Payout"; U.S. Pat. No. 7,666,093, entitled "Gaming Method and Device Involving Progressive Wagers"; U.S. Pat. No. 7,780,523, entitled "Server Based Gaming System Having Multiple Progressive Awards"; and U.S. Pat. No. 8,337,298, entitled "Gaming Device Having Multiple Different Types of Progressive Awards."

As generally noted above, in addition to providing winning credits or other awards for one or more plays of the primary game(s), in various embodiments the gaming system provides credits or other awards for one or more plays of one or more secondary games. The secondary game typically enables an award to be obtained addition to any award obtained through play of the primary game(s). The secondary game(s) typically produces a higher level of player excitement than the primary game(s) because the secondary game(s) provides a greater expectation of winning than the primary game(s) and is accompanied with more attractive or unusual features than the primary game(s). The secondary game(s) may be any type of suitable game, either similar to or completely different from the primary game.

In various embodiments, the gaming system automatically provides or initiates the secondary game upon the occurrence of a triggering event or the satisfaction of a qualifying condition. In other embodiments, the gaming system initiates the secondary game upon the occurrence of the triggering event or the satisfaction of the qualifying condition and upon receipt of an initiation input. In certain embodiments, the triggering event or qualifying condition is a selected outcome in the primary game(s) or a particular arrangement of one or more indicia on a display device for a play of the primary game(s), such as a "BONUS" symbol appearing on three adjacent reels along a payline following a spin of the reels for a play of the primary game. In other embodiments, the triggering event or qualifying condition occurs based on a certain amount of game play (such as number of games, number of credits, amount of time) being exceeded, or based on a specified number of points being earned during game play. Any suitable triggering event or qualifying condition or any suitable combination of a plurality of different triggering events or qualifying conditions may be employed.

In other embodiments, at least one processor of the gaming system randomly determines when to provide one or more plays of one or more secondary games. In one such embodiment, no apparent reason is provided for providing the secondary game. In this embodiment, qualifying for a secondary game is not triggered by the occurrence of an event in any primary game or based specifically on any of the plays of any primary game. That is, qualification is provided without any explanation or, alternatively, with a simple explanation. In another such embodiment, the gaming system determines qualification for a secondary game at least partially based on a game triggered or symbol triggered event, such as at least partially based on play of a primary game.

In various embodiments, after qualification for a secondary game has been determined, the secondary game participation may be enhanced through continued play on the primary game. Thus, in certain embodiments, for each secondary game qualifying event, such as a secondary game symbol, which is obtained, a given number of secondary game wagering points or credits is accumulated in a "secondary game meter" configured to accrue the secondary game wagering credits or entries toward eventual participation in the secondary game. In one such embodiment, the occurrence of multiple such secondary game qualifying events in the primary game results in an arithmetic or exponential increase in the number of secondary game wagering credits awarded. In another such embodiment, any extra secondary game wagering credits may be redeemed during the secondary game to extend play of the secondary game.

In certain embodiments, no separate entry fee or buy-in for the secondary game is required. That is, entry into the secondary game cannot be purchased; rather, in these embodiments' entry must be won or earned through play of the primary game, thereby encouraging play of the primary game. In other embodiments, qualification for the secondary game is accomplished through a simple "buy-in." For example, qualification through other specified activities is unsuccessful, payment of a fee or placement of an additional wager "buys-in" to the secondary game. In certain embodiments, a separate side wager must be placed on the secondary game, or a wager of a designated amount must be placed on the primary game to enable qualification for the secondary game. In these embodiments, the secondary game triggering event must occur and the side wager (or designated primary game wager amount) must have been placed for the secondary game to trigger.

In various embodiments in which the gaming system includes a plurality of EGMs, the EGMs are configured to communicate with one another to provide a group gaming environment. In certain such embodiments, the EGMs enable players of those EGMs to work in conjunction with one another, such as by enabling the players to play together as a team or group, to win one or more awards. In other such embodiments, the EGMs enable players of those EGMs to compete against one another for one or more awards. In one such embodiment, the EGMs enable the players of those EGMs to participate in one or more gaming tournaments for one or more awards. Examples of group gaming systems are described in U.S. Pat. No. 8,070,583, entitled "Server Based Gaming System and Method for Selectively Providing One or More Different Tournaments"; U.S. Pat. No. 8,500,548, entitled "Gaming System and Method for Providing Team Progressive Awards"; and U.S. Pat. No. 8,562,423, entitled "Method and Apparatus for Rewarding Multiple Game Players for a Single Win."

In various embodiments, the gaming system includes one or more player tracking systems. Such player tracking systems enable operators of the gaming system (such as casinos or other gaming establishments) to recognize the value of customer loyalty by identifying frequent customers and rewarding them for their patronage. Such a player tracking system is configured to track a player's gaming activity. In one such embodiment, the player tracking system does so through the use of player tracking cards. In this embodiment, a player is issued a player identification card that has an encoded player identification number that uniquely identifies the player. When the player's playing tracking card is inserted into a card reader of the gaming system to begin a gaming session, the card reader reads the player identification number off the player tracking card to identify the player. The gaming system timely tracks any suitable information or data relating to the identified player's gaming session. The gaming system also timely tracks when the player tracking card is removed to conclude play for that gaming session. In another embodiment, rather than requiring insertion of a player tracking card into the card reader, the gaming system utilizes one or more portable devices, such as a mobile phone, a radio frequency identification tag, or any other suitable wireless device, to track when a gaming session begins and ends. In another embodiment, the gaming system utilizes any suitable biometric technology or ticket technology to track when a gaming session begins and ends.

In such embodiments, during one or more gaming sessions, the gaming system tracks any suitable information or data, such as any amounts wagered, average wager amounts, and/or the time at which these wagers are placed. In different

embodiments, for one or more players, the player tracking system includes the player's account number, the player's card number, the player's first name, the player's surname, the player's preferred name, the player's player tracking ranking, any promotion status associated with the player's player tracking card, the player's address, the player's birthday, the player's anniversary, the player's recent gaming sessions, or any other suitable data. In various embodiments, such tracked information and/or any suitable feature associated with the player tracking system is displayed on a player tracking display. In various embodiments, such tracked information and/or any suitable feature associated with the player tracking system is displayed via one or more service windows that are displayed on the central display device and/or the upper display device. Examples of player tracking systems are described in U.S. Pat. No. 6,722,985, entitled "Universal Player Tracking System"; U.S. Pat. No. 6,908,387, entitled "Player Tracking Communication Mechanisms in a Gaming Machine"; U.S. Pat. No. 7,311,605, entitled "Player Tracking Assembly for Complete Patron Tracking for Both Gaming and Non-Gaming Casino Activity"; U.S. Pat. No. 7,611,411, entitled "Player Tracking Instruments Having Multiple Communication Modes"; U.S. Pat. No. 7,617,151, entitled "Alternative Player Tracking Techniques"; and U.S. Pat. No. 8,057,298, entitled "Virtual Player Tracking and Related Services."

Web-Based Gaming

In various embodiments, the gaming system includes one or more servers configured to communicate with a personal gaming device—such as a smartphone, a tablet computer, a desktop computer, or a laptop computer—to enable web-based game play using the personal gaming device. In various embodiments, the player must first access a gaming website via an Internet browser of the personal gaming device or execute an application (commonly called an "app") installed on the personal gaming device before the player can use the personal gaming device to participate in web-based game play. In certain embodiments, the one or more servers and the personal gaming device operate in a thin-client environment. In these embodiments, the personal gaming device receives inputs via one or more input devices (such as a touch screen and/or physical buttons), the personal gaming device sends the received inputs to the one or more servers, the one or more servers make various determinations based on the inputs and determine content to be displayed (such as a randomly determined game outcome and corresponding award), the one or more servers send the content to the personal gaming device, and the personal gaming device displays the content.

In certain such embodiments, the one or more servers must identify the player before enabling game play on the personal gaming device (or, in some embodiments, before enabling monetary wager-based game play on the personal gaming device). In these embodiments, the player must identify herself to the one or more servers, such as by inputting the player's unique username and password combination, providing an input to a biometric sensor (e.g., a fingerprint sensor, a retinal sensor, a voice sensor, or a facial-recognition sensor), or providing any other suitable information.

Once identified, the one or more servers enable the player to establish an account balance from which the player can draw credits usable to wager on plays of a game. In certain embodiments, the one or more servers enable the player to initiate an electronic funds transfer to transfer funds from a

bank account to the player's account balance. In other embodiments, the one or more servers enable the player to make a payment using the player's credit card, debit card, or other suitable device to add money to the player's account balance. In other embodiments, the one or more servers enable the player to add money to the player's account balance via a peer-to-peer type application, such as PayPal or Venmo. The one or more servers also enable the player to cash out the player's account balance (or part of it) in any suitable manner, such as via an electronic funds transfer, by initiating creation of a paper check that is mailed to the player, or by initiating printing of a voucher at a kiosk in a gaming establishment.

In certain embodiments, the one or more servers include a payment server that handles establishing and cashing out players' account balances and a separate game server configured to determine the outcome and any associated award for a play of a game. In these embodiments, the game server is configured to communicate with the personal gaming device and the payment device, and the personal gaming device and the payment device are not configured to directly communicate with one another. In these embodiments, when the game server receives data representing a request to start a play of a game at a desired wager, the game server sends data representing the desired wager to the payment server. The payment server determines whether the player's account balance can cover the desired wager (i.e., includes a monetary balance at least equal to the desired wager).

If the payment server determines that the player's account balance cannot cover the desired wager, the payment server notifies the game server, which then instructs the personal gaming device to display a suitable notification to the player that the player's account balance is too low to place the desired wager. If the payment server determines that the player's account balance can cover the desired wager, the payment server deducts the desired wager from the account balance and notifies the game server. The game server then determines an outcome and any associated award for the play of the game. The game server notifies the payment server of any nonzero award, and the payment server increases the player's account balance by the nonzero award. The game server sends data representing the outcome and any award to the personal gaming device, which displays the outcome and any award.

In certain embodiments, the one or more servers enable web-based game play using a personal gaming device only if the personal gaming device satisfies one or more jurisdictional requirements. In one embodiment, the one or more servers enable web-based game play using the personal gaming device only if the personal gaming device is located within a designated geographic area (such as within certain state or county lines or within the boundaries of a gaming establishment). In this embodiment, the geolocation module of the personal gaming device determines the location of the personal gaming device and sends the location to the one or more servers, which determine whether the personal gaming device is located within the designated geographic area. In various embodiments, the one or more servers enable non-monetary wager-based game play if the personal gaming device is located outside of the designated geographic area.

In various embodiments, the gaming system includes an EGM configured to communicate with a personal gaming device—such as a smartphone, a tablet computer, a desktop computer, or a laptop computer—to enable tethered mobile game play using the personal gaming device. Generally, in these embodiments, the EGM establishes communication with the personal gaming device and enables the player to

play games on the EGM remotely via the personal gaming device. In certain embodiments, the gaming system includes a geo-fence system that enables tethered game play within a particular geographic area but not outside of that geographic area. Examples of tethering an EGM to a personal gaming device and geo-fencing are described in U.S. Patent Appl. Pub. No. 2013/0267324, entitled "Remote Gaming Method Allowing Temporary Inactivation Without Terminating Playing Session Due to Game Inactivity."

Social Network Integration

In certain embodiments, the gaming system is configured to communicate with a social network server that hosts or partially hosts a social networking website via a data network (such as the Internet) to integrate a player's gaming experience with the player's social networking account. This enables the gaming system to send certain information to the social network server that the social network server can use to create content (such as text, an image, and/or a video) and post it to the player's wall, newsfeed, or similar area of the social networking website accessible by the player's connections (and in certain cases the public) such that the player's connections can view that information. This also enables the gaming system to receive certain information from the social network server, such as the player's likes or dislikes or the player's list of connections. In certain embodiments, the gaming system enables the player to link the player's player account to the player's social networking account(s). This enables the gaming system to, once it identifies the player and initiates a gaming session (such as via the player logging in to a website (or an application) on the player's personal gaming device or via the player inserting the player's player tracking card into an EGM), link that gaming session to the player's social networking account(s). In other embodiments, the gaming system enables the player to link the player's social networking account(s) to individual gaming sessions when desired by providing the required login information.

For instance, in one embodiment, if a player wins a particular award (e.g., a progressive award or a jackpot award) or an award that exceeds a certain threshold (e.g., an award exceeding \$1,000), the gaming system sends information about the award to the social network server to enable the server to create associated content (such as a screenshot of the outcome and associated award) and to post that content to the player's wall (or other suitable area) of the social networking website for the player's connections to see (and to entice them to play). In another embodiment, if a player joins a multiplayer game and there is another seat available, the gaming system sends that information to the social network sever to enable the server to create associated content (such as text indicating a vacancy for that particular game) and to post that content to the player's wall (or other suitable area) of the social networking website for the player's connections to see (and to entice them to fill the vacancy). In another embodiment, if the player consents, the gaming system sends advertisement information or offer information to the social network server to enable the social network server to create associated content (such as text or an image reflecting an advertisement and/or an offer) and to post that content to the player's wall (or other suitable area) of the social networking website for the player's connections to see. In another embodiment, the gaming system enables the player to recommend a game to the player's connections

by posting a recommendation to the player's wall (or other suitable area) of the social networking website.

Differentiating Certain Gaming Systems from General Purpose Computing Devices

Certain of the gaming systems described herein, such as EGMs located in a casino or another gaming establishment, include certain components and/or are configured to operate in certain manners that differentiate these systems from general purpose computing devices, i.e., certain personal gaming devices such as desktop computers and laptop computers.

For instance, EGMs are highly regulated to ensure fairness, and, in many cases, EGMs are configured to award monetary awards up to multiple millions of dollars. To satisfy security and regulatory requirements in a gaming environment, hardware and/or software architectures are implemented in EGMs that differ significantly from those of general purpose computing devices. For purposes of illustration, a description of EGMs relative to general purpose computing devices and some examples of these additional (or different) hardware and/or software architectures found in EGMs are described below.

At first glance, one might think that adapting general purpose computing device technologies to the gaming industry and EGMs would be a simple proposition because both general purpose computing devices and EGMs employ processors that control a variety of devices. However, due to at least: (1) the regulatory requirements placed on EGMs, (2) the harsh environment in which EGMs operate, (3) security requirements, and (4) fault tolerance requirements, adapting general purpose computing device technologies to EGMs can be quite difficult. Further, techniques and methods for solving a problem in the general purpose computing device industry, such as device compatibility and connectivity issues, might not be adequate in the gaming industry. For instance, a fault or a weakness tolerated in a general purpose computing device, such as security holes in software or frequent crashes, is not tolerated in an EGM because in an EGM these faults can lead to a direct loss of funds from the EGM, such as stolen cash or loss of revenue when the EGM is not operating properly or when the random outcome determination is manipulated.

Certain differences between general purpose computing devices and EGMs are described below. A first difference between EGMs and general purpose computing devices is that EGMs are state-based systems. A state-based system stores and maintains its current state in a non-volatile memory such that, in the event of a power failure or other malfunction, the state-based system can return to that state when the power is restored, or the malfunction is remedied. For instance, for a state-based EGM, if the EGM displays an award for a game of chance but the power to the EGM fails before the EGM provides the award to the player, the EGM stores the pre-power failure state in a non-volatile memory, returns to that state upon restoration of power, and provides the award to the player. This requirement affects the software and hardware design on EGMs. General purpose computing devices are not state-based machines, and a majority of data is usually lost when a malfunction occurs on a general purpose computing device.

A second difference between EGMs and general purpose computing devices is that, for regulatory purposes, the software on the EGM utilized to operate the EGM has been designed to be static and monolithic to prevent cheating by the operator of the EGM. For instance, one solution that has

been employed in the gaming industry to prevent cheating and to satisfy regulatory requirements has been to manufacture an EGM that can use a proprietary processor running instructions to provide the game of chance from an EPROM or other form of non-volatile memory. The coding instructions on the EPROM are static (non-changeable) and must be approved by a gaming regulators in a particular jurisdiction and installed in the presence of a person representing the gaming jurisdiction. Any changes to any part of the software required to generate the game of chance, such as adding a new device driver used to operate a device during generation of the game of chance, can require burning a new EPROM approved by the gaming jurisdiction and reinstalling the new EPROM on the EGM in the presence of a gaming regulator. Regardless of whether the EPROM solution is used, to gain approval in most gaming jurisdictions, an EGM must demonstrate sufficient safeguards that prevent an operator or a player of an EGM from manipulating the EGM's hardware and software in a manner that gives him an unfair, and in some cases illegal, advantage.

A third difference between EGMs and general purpose computing devices is authentication-EGMs storing code are configured to authenticate the code to determine if the code is unaltered before executing the code. If the code has been altered, the EGM prevents the code from being executed. The code authentication requirements in the gaming industry affect both hardware and software designs on EGMs. Certain EGMs use hash functions to authenticate code. For instance, one EGM stores game program code, a hash function, and an authentication hash (which may be encrypted). Before executing the game program code, the EGM hashes the game program code using the hash function to obtain a result hash and compares the result hash to the authentication hash. If the result hash matches the authentication hash, the EGM determines that the game program code is valid and executes the game program code. If the result hash does not match the authentication hash, the EGM determines that the game program code has been altered (i.e., may have been tampered with) and prevents execution of the game program code. Examples of EGM code authentication are described in U.S. Pat. No. 6,962,530, entitled "Authentication in a Secure Computerized Gaming System"; U.S. Pat. No. 7,043,641, entitled "Encryption in a Secure Computerized Gaming System"; U.S. Pat. No. 7,201,662, entitled "Method and Apparatus for Software Authentication"; and U.S. Pat. No. 8,627,097, entitled "System and Method Enabling Parallel Processing of Hash Functions Using Authentication Check-point Hashes."

A fourth difference between EGMs and general purpose computing devices is that EGMs have unique peripheral device requirements that differ from those of a general purpose computing device, such as peripheral device security requirements not usually addressed by general purpose computing devices. For instance, monetary devices, such as coin dispensers, bill validators, and ticket printers and computing devices that are used to govern the input and output of cash or other items having monetary value (such as tickets) to and from an EGM have security requirements that are not typically addressed in general purpose computing devices. Therefore, many general purpose computing device techniques and methods developed to facilitate device connectivity and device compatibility do not address the emphasis placed on security in the gaming industry.

To address some of the issues described above, a number of hardware/software components and architectures are utilized in EGMs that are not typically found in general purpose computing devices. These hardware/software com-

ponents and architectures, as described below in more detail, include but are not limited to watchdog timers, voltage monitoring systems, state-based software architecture and supporting hardware, specialized communication interfaces, security monitoring, and trusted memory.

Certain EGMs use a watchdog timer to provide a software failure detection mechanism. In a normally-operating EGM, the operating software periodically accesses control registers in the watchdog timer subsystem to "re-trigger" the watchdog. Should the operating software fail to access the control registers within a preset timeframe, the watchdog timer will timeout and generate a system reset. Typical watchdog timer circuits include a loadable timeout counter register to enable the operating software to set the timeout interval within a certain range of time. A differentiating feature of some circuits is that the operating software cannot completely disable the function of the watchdog timer. In other words, the watchdog timer always functions from the time power is applied to the board.

Certain EGMs use several power supply voltages to operate portions of the computer circuitry. These can be generated in a central power supply or locally on the computer board. If any of these voltages falls out of the tolerance limits of the circuitry they power, unpredictable operation of the EGM may result. Though most modern general purpose computing devices include voltage monitoring circuitry, these types of circuits only report voltage status to the operating software. Out of tolerance voltages can cause software malfunction, creating a potential uncontrolled condition in the general purpose computing device. Certain EGMs have power supplies with relatively tighter voltage margins than that required by the operating circuitry. In addition, the voltage monitoring circuitry implemented in certain EGMs typically has two thresholds of control. The first threshold generates a software event that can be detected by the operating software and an error condition then generated. This threshold is triggered when a power supply voltage falls out of the tolerance range of the power supply, but is still within the operating range of the circuitry. The second threshold is set when a power supply voltage falls out of the operating tolerance of the circuitry. In this case, the circuitry generates a reset, halting operation of the EGM.

As described above, certain EGMs are state-based machines. Different functions of the game provided by the EGM (e.g., bet, play, result, points in the graphical presentation, etc.) may be defined as a state. When the EGM moves a game from one state to another, the EGM stores critical data regarding the game software in a custom non-volatile memory subsystem. This ensures that the player's wager and credits are preserved and to minimize potential disputes in the event of a malfunction on the EGM. In general, the EGM does not advance from a first state to a second state until critical information that enables the first state to be reconstructed has been stored. This feature enables the EGM to recover operation to the current state of play in the event of a malfunction, loss of power, etc. that occurred just before the malfunction. In at least one embodiment, the EGM is configured to store such critical information using atomic transactions.

Generally, an atomic operation in computer science refers to a set of operations that can be combined so that they appear to the rest of the system to be a single operation with only two possible outcomes: success or failure. As related to data storage, an atomic transaction may be characterized as series of database operations which either all occur, or all do

not occur. A guarantee of atomicity prevents updates to the database occurring only partially, which can result in data corruption.

To ensure the success of atomic transactions relating to critical information to be stored in the EGM memory before a failure event (e.g., malfunction, loss of power, etc.), memory that includes one or more of the following criteria be used: direct memory access capability; data read/write access characteristics (such as at least 5.08 Mbytes/see (Read) and/or at least 38.0 Mbytes/see (Write)). Memory devices that meet or exceed the above criteria may be referred to as "fault-tolerant" memory devices.

Typically, battery-backed RAM devices may be configured to function as fault-tolerant devices according to the above criteria, whereas flash RAM and/or disk drive memory are typically not configurable to function as fault-tolerant devices according to the above criteria. Accordingly, battery-backed RAM devices are typically used to preserve EGM critical data, although other types of non-volatile memory devices may be employed. These memory devices are typically not used in typical general purpose computing devices.

Thus, in at least one embodiment, the EGM is configured to store critical information in fault-tolerant memory (e.g., battery-backed RAM devices) using atomic transactions. Further, in at least one embodiment, the fault-tolerant memory is able to successfully complete all desired atomic transactions (e.g., relating to the storage of EGM critical information) within a time period of 200 milliseconds or less. In at least one embodiment, the time period of 200 milliseconds represents a maximum amount of time for which sufficient power may be available to the various EGM components after a power outage event has occurred at the EGM.

As described previously, the EGM may not advance from a first state to a second state until critical information that enables the first state to be reconstructed has been atomically stored. After the state of the EGM is restored during the play of a game of chance, game play may resume, and the game may be completed in a manner that is no different than if the malfunction had not occurred. Thus, for example, when a malfunction occurs during a game of chance, the EGM may be restored to a state in the game of chance just before when the malfunction occurred. The restored state may include metering information and graphical information that was displayed on the EGM in the state before the malfunction. For example, when the malfunction occurs during the play of a card game after the cards have been dealt, the EGM may be restored with the cards that were previously displayed as part of the card game. As another example, a bonus game may be triggered during the play of a game of chance in which a player is required to make a number of selections on a video display screen. When a malfunction has occurred after the player has made one or more selections, the EGM may be restored to a state that shows the graphical presentation just before the malfunction including an indication of selections that have already been made by the player. In general, the EGM may be restored to any state in a plurality of states that occur in the game of chance that occurs while the game of chance is played or to states that occur between the play of a game of chance.

Game history information regarding previous games played such as an amount wagered, the outcome of the game, and the like may also be stored in a non-volatile memory device. The information stored in the non-volatile memory may be detailed enough to reconstruct a portion of

the graphical presentation that was previously presented on the EGM and the state of the EGM (e.g., credits) at the time the game of chance was played. The game history information may be utilized in the event of a dispute. For example, a player may decide that in a previous game of chance that they did not receive credit for an award that they believed they won. The game history information may be used to reconstruct the state of the EGM before, during, and/or after the disputed game to demonstrate whether the player was correct or not in the player's assertion. Examples of a state-based EGM, recovery from malfunctions, and game history are described in U.S. Pat. No. 6,804,763, entitled "High Performance Battery Backed RAM Interface"; U.S. Pat. No. 6,863,608, entitled "Frame Capture of Actual Game Play"; U.S. Pat. No. 7,111,141, entitled "Dynamic NV-RAM"; and U.S. Pat. No. 7,384,339, entitled, "Frame Capture of Actual Game Play."

Another feature of EGMs is that they often include unique interfaces, including serial interfaces, to connect to specific subsystems internal and external to the EGM. The serial devices may have electrical interface requirements that differ from the "standard" EIA serial interfaces provided by general purpose computing devices. These interfaces may include, for example, Fiber Optic Serial, optically coupled serial interfaces, current loop style serial interfaces, etc. In addition, to conserve serial interfaces internally in the EGM, serial devices may be connected in a shared, daisy-chain fashion in which multiple peripheral devices are connected to a single serial channel.

The serial interfaces may be used to transmit information using communication protocols that are unique to the gaming industry. For example, IGT's Netplex is a proprietary communication protocol used for serial communication between EGMs. As another example, SAS is a communication protocol used to transmit information, such as metering information, from an EGM to a remote device. Often SAS is used in conjunction with a player tracking system.

Certain EGMs may alternatively be treated as peripheral devices to a casino communication controller and connected in a shared daisy chain fashion to a single serial interface. In both cases, the peripheral devices are assigned device addresses. If so, the serial controller circuitry must implement a method to generate or detect unique device addresses. General purpose computing device serial ports are not able to do this.

Security monitoring circuits detect intrusion into an EGM by monitoring security switches attached to access doors in the EGM cabinet. Access violations result in suspension of game play and can trigger additional security operations to preserve the current state of game play. These circuits also function when power is off by use of a battery backup. In power-off operation, these circuits continue to monitor the access doors of the EGM. When power is restored, the EGM can determine whether any security violations occurred while power was off, e.g., via software for reading status registers. This can trigger event log entries and further data authentication operations by the EGM software.

Trusted memory devices and/or trusted memory sources are included in an EGM to ensure the authenticity of the software that may be stored on less secure memory subsystems, such as mass storage devices. Trusted memory devices and controlling circuitry are typically designed to not enable modification of the code and data stored in the memory device while the memory device is installed in the EGM. The code and data stored in these devices may include authentication algorithms, random number generators, authentication keys, operating system kernels, etc. The pur-

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pose of these trusted memory devices is to provide gaming regulatory authorities a root trusted authority within the computing environment of the EGM that can be tracked and verified as original. This may be accomplished via removal of the trusted memory device from the EGM computer and verification of the secure memory device contents is a separate third party verification device. Once the trusted memory device is verified as authentic, and based on the approval of the verification algorithms included in the trusted device, the EGM is enabled to verify the authenticity of additional code and data that may be located in the gaming computer assembly, such as code and data stored on hard disk drives. Examples of trusted memory devices are described in U.S. Pat. No. 6,685,567, entitled "Process Verification."

In at least one embodiment, at least a portion of the trusted memory devices/sources may correspond to memory that cannot easily be altered (e.g., "unalterable memory") such as EPROMS, PROMS, Bios, Extended Bios, and/or other memory sources that are able to be configured, verified, and/or authenticated (e.g., for authenticity) in a secure and controlled manner.

According to one embodiment, when a trusted information source is in communication with a remote device via a network, the remote device may employ a verification scheme to verify the identity of the trusted information source. For example, the trusted information source and the remote device may exchange information using public and private encryption keys to verify each other's identities. In another embodiment, the remote device and the trusted information source may engage in methods using zero knowledge proofs to authenticate each of their respective identities.

EGMs storing trusted information may utilize apparatuses or methods to detect and prevent tampering. For instance, trusted information stored in a trusted memory device may be encrypted to prevent its misuse. In addition, the trusted memory device may be secured behind a locked door. Further, one or more sensors may be coupled to the memory device to detect tampering with the memory device and provide some record of the tampering. In yet another example, the memory device storing trusted information might be designed to detect tampering attempts and clear or erase itself when an attempt at tampering has been detected. Examples of trusted memory devices/sources are described in U.S. Pat. No. 7,515,718, entitled "Secured Virtual Network in a Gaming Environment."

Mass storage devices used in a general purpose computing devices typically enable code and data to be read from and written to the mass storage device. In a gaming environment, modification of the gaming code stored on a mass storage device is strictly controlled and would only be enabled under specific maintenance type events with electronic and physical enablers required. Though this level of security could be provided by software, EGMs that include mass storage devices include hardware level mass storage data protection circuitry that operates at the circuit level to monitor attempts to modify data on the mass storage device and will generate both software and hardware error triggers should a data modification be attempted without the proper electronic and physical enablers being present. Examples of using a mass storage device are described in U.S. Pat. No. 6,149,522, entitled "Method of Authenticating Game Data Sets in an Electronic Casino Gaming System."

Various changes and modifications to the present embodiments described herein will be apparent to those skilled in the art. Such changes and modifications can be made without

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departing from the spirit and scope of the present subject matter and without diminishing its intended advantages. It is therefore intended that such changes and modifications be covered by the appended claims.

What is claimed is:

1. A gaming system comprising:

a processor; and

a memory device that stores a plurality of instructions that, when executed by the processor, cause the processor to:

cause a display, by a display device, of a first sequence triggering event,

cause a display, by the display device, of a first sequence comprising plays of a game,

cause a display, by the display device, of a plurality of accumulated symbol displays that are generated during the first sequence, each accumulated symbol display comprising a quantity of different symbols, wherein two of the accumulated symbol displays comprises different quantities of symbols,

cause a display, by the display device, of the quantity of different symbols of one of the accumulated symbol displays changing during the first sequence,

cause a display, by the display device, of two of the accumulated symbol displays changing into a combined accumulated symbol display during the first sequence,

cause a display, by the display device, of a triggering event for a second sequence, and

cause a display, by the display device, of the second sequence wherein one of the accumulated symbol displays remaining after the triggering event for the second sequence occurs is activated and displays one of the symbols on said accumulated symbol display.

2. The gaming system of claim 1, wherein the plurality of instructions, when executed by the processor, cause the processor to cause a display, by the display device, of the quantity of accumulated symbol displays changing during the first sequence.

3. The gaming system of claim 1, wherein the plurality of instructions, when executed by the processor, cause the processor to cause a display, by the display device, of an additional accumulated symbol display added to the displayed accumulated symbol displays during the first sequence.

4. The gaming system of claim 1, wherein the plurality of instructions, when executed by the processor, cause the processor to cause a display, by the display device, of one of the accumulated symbol displays removed from the displayed accumulated symbol displays during the first sequence.

5. The gaming system of claim 1, wherein the plurality of instructions, when executed by the processor, cause the processor to cause a display, by the display device, of the combined accumulated symbol display comprising a sum of the quantities of symbols of the two accumulated symbol displays that change into the combined accumulated symbol display.

6. The gaming system of claim 1, wherein the plurality of instructions, when executed by the processor, cause the processor to cause a display, by the display device, of one of the accumulated symbol displays being split into two accumulated symbol displays during the first sequence.

7. The gaming system of claim 6, wherein the plurality of instructions, when executed by the processor, cause the processor to cause a display, by the display device, of the two accumulated symbol displays that are split from the one

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accumulated symbol display comprise a total quantity of symbols equal to the quantity of symbols of the one accumulated symbol display.

8. The gaming system of claim 1, wherein the plurality of instructions, when executed by the processor, cause the processor to cause a display, by the display device, of the second sequence wherein all of the accumulated symbol displays are activated and each accumulated symbol display displays one of the symbols of said accumulated symbol display.

9. The gaming system of claim 1, wherein the plurality of instructions, when executed by the processor, cause the processor to cause a display, by the display device, of an initial accumulated symbol display in association with the display of the first sequence triggering event.

10. The gaming system of claim 1, wherein the plurality of instructions, when executed by the processor, cause the processor to cause a display, by the display device, of each of an initial quantity of accumulated symbol displays in association with the display of the first sequence triggering event, and for each initial accumulated symbol display a quantity of symbols on said accumulated symbol display.

11. The gaming system of claim 1, wherein the plurality of instructions, when executed by the processor, cause the processor to cause a display, by the display device, for each of a plurality of the plays of the game of the first sequence, of a randomly determined plurality of symbols, wherein one of the plurality of symbols causes a change to the accumulated symbol displays for the first sequence.

12. A gaming system comprising:

a processor; and

a memory device that stores a plurality of instructions that, when executed by the processor, cause the processor to:

cause a display, by a display device, of a first sequence triggering event,

cause a display, by the display device, of each of an initial quantity of accumulated symbol displays after the display of the first sequence triggering event, and for each said accumulated symbol display a quantity of symbols on said accumulated symbol display, the quantity being at least one,

cause a display, by the display device, of a first sequence comprising plays of a game,

cause a display, by the display device, of the accumulated symbol displays changing during the first sequence, and which comprises a display, by the display device, of the quantity of different symbols on one of the accumulated symbol displays changing during the first sequence, and which comprises a display, by the display device, of the quantity of the accumulated symbol displays changing during the first sequence,

cause a display, by the display device, of one of the accumulated symbol displays being split into two accumulated symbol displays during the first sequence,

cause a display, by the display device, of a triggering event for a second sequence, and

cause a display, by the display device, of the second sequence wherein each of the accumulated symbol displays remaining after the triggering event for the second sequence occurs is activated and displays one of the symbols on said accumulated symbol display.

13. The gaming system of claim 12, wherein the plurality of instructions, when executed by the processor, cause the processor to cause a display, by the display device, of an

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additional accumulated symbol display to be added to the displayed accumulated symbol displays during the first sequence.

14. The gaming system of claim 12, wherein the plurality of instructions, when executed by the processor, cause the processor to cause a display, by the display device, of one of the accumulated symbol displays to be removed from the displayed accumulated symbol displays during the first sequence.

15. A gaming system comprising:

a processor; and

a memory device that stores a plurality of instructions that, when executed by the processor, cause the processor to:

cause a display, by a display device, of a first sequence triggering event,

cause a display, by the display device, of a first sequence comprising plays of a game,

cause a display, by the display device, of a plurality of accumulated symbol displays that are displayed during the first sequence, each of the accumulated symbol displays comprising a quantity of different symbols, and two of the accumulated symbol displays comprising different quantities of symbols,

cause a display, by the display device, of two of the accumulated symbol displays being formed into a combined accumulated symbol display during the first sequence, wherein the combined accumulated symbol display comprises a sum of the quantities of symbols of the two accumulated symbol displays that are formed into the combined accumulated symbol display,

cause a display, by the display device, of one of the accumulated symbol displays being split into two accumulated symbol displays during the first sequence, wherein the two accumulated symbol displays that are split from the one accumulated symbol display comprise a total quantity of symbols equal to the quantity of symbols of said one accumulated symbol display,

cause a display, by the display device, of a triggering event for a second sequence, and

cause a display, by the display device, of the second sequence wherein each of the accumulated symbol displays remaining after the triggering event for the second sequence occurs is activated and displays one of the symbols on said accumulated symbol display.

16. The gaming system of claim 15, wherein the plurality of instructions, when executed by the processor, cause the processor to cause a display, by the display device, of an initial accumulated symbol display in association with the display of the first sequence triggering event.

17. The gaming system of claim 15, wherein the plurality of instructions, when executed by the processor, cause the processor to cause a display, by the display device, of each of an initial quantity of accumulated symbol displays in association with the display of the first sequence triggering event, and for each of said accumulated symbol display a quantity of symbols on said accumulated symbol display.

18. The gaming system of claim 15, wherein the plurality of instructions, when executed by the processor, cause the processor to cause a display, by the display device, for each of a plurality of plays of the game of the first sequence, of a randomly determined plurality of symbols, wherein one of the plurality of symbols causes a change to the accumulated symbol displays during the first sequence.

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19. The gaming system of claim **15**, wherein the plurality of instructions, when executed by the processor, cause the processor to cause a display, by the display device, of an additional accumulated symbol display to be added to the accumulated symbol displays during the first sequence. 5

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